

Emotion (2003/2007)

Software System

Installation of SOMARIS/5 VB42B

This document is valid for:
SOMATOM Emotion / Emotion Duo
SOMATOM Emotion 6-slice configuration
SOMATOM Emotion 16-slice configuration
SOMATOM Sensation 16
SOMATOM Sensation 40
SOMATOM Sensation 64
SOMATOM Sensation Open

Document Version

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1.1 Definitions and Abbreviations

Abbreviations	Definitions
ACR-NEMA	American College of Radiology - National Electrical Manufacturers Association
BIOS	Basic Input Output System, lowest PC configuration settings
AET	Application Entity Title - Name of DICOM service participating in the communication either as a user (SCU) or a provider (SCP)
CD	Compact Disk
CDR	Compact Disk Recordable
CT	Computed Tomography
DICOM	Digital Image Communication in Medicine (ACR-NEMA standard)
DHCP	Dynamic Host Configuration Protocol - procedure for assigning a numeric IP address to nodes in which blocks of addresses are leased to clients on a first-come, first-serve basis. Typically, DHCP is combined with WINS. This means that an IP address is both dynamically assigned and automatically associated with a domain name. This can be combined with DNS, resulting in a dynamically assigned numeric IP address that is automatically associated with both NetBIOS and Internet names.
DNS	Domain Name Services - address resolution procedure that associates numeric IP addresses with Internet-style names
FTP	File Transfer Protocol - a client-server application that transfers files from one computer to another
Gateway	A device that controls the traffic between two networks using protocols above the network layer
ICS	Image Control System - control computer for the entire CT scanner, with interface to the acquisition system; syngo CT Acquisition Workplace.
IES	Image Evaluation System; syngo CT workplace
IMS	Image Measurement System, consisting of ICS, IRS, and console
IRS	Image Reconstruction System - reconstruction computer
LAN	Local Area Network (here, always Ethernet)
LMHOSTS	Address resolution procedure which statically associates numeric IP addresses with NetBIOS names. The LMHOSTS file is checked to see if the name is being addressed in this file.
NetBIOS	Network Basic Input Output System - a collection of network services and programming interfaces. NetBIOS services run on network protocols such as TCP/IP.

Abbreviations	Definitions
NIC	Network Interface Card - an expansion card that provides a physical connection to the network
NT	"New Technology" Operating System - trademark of Microsoft, Inc.
OI	Operating Instructions
PB	Patient browser
PPP	Point-to-Point Protocol - a data link and network protocol designed for use over serial connections (such as dial-up modems)
RAS	Remote Access Services: a Windows NT service that supports dial-in and dial-out connections using modems and .
Router	A multiport computer or piece of dedicated hardware that examines network layer addresses of packets on a network and retransmits them on a different network, if necessary
SCU	Service Class User
SCP	Service Class Provider
TCP/IP	Transport Control Protocol/Internet Protocol - the most widely used standard for networking
UI	User Interface
USC	Uptime Service Center
WINS	Windows Internet Name Service - address resolution procedure that dynamically associates numeric IP addresses with NetBIOS names. It is possible to combine DNS and WINS.
XP	"eXPerience" Operating System - trademark of Microsoft, Inc.

1.2 Safety Information

Only qualified and trained personnel are allowed to perform CT system installation, service, maintenance, and quality assurance.

To avoid risk of injury to personnel or damage to the system, the safety instructions must be read and observed as described in **General Safety Notes (TD00-000.860.01.xx.xx)** and Product-specific Safety Notes (CT02-025.860.02.xx.xx).

The product-specific safety notes contain important information about:

- General safety information regarding the gantry, patient table, and laser products.
- Attachment of screws and nuts, adjustment instructions for torque wrench, and the use of Loctite.
- Safety advice (hazard keys)

1.3 Overview

ICS = syngo Acquisition Workplace

In the standard configuration, the ICS is the main console for interchangeable scanning and imaging functions. Only the IRS3 of Emotion 16 systems is equipped with a switchable keyboard, mouse, and monitor. Since August 2006, the IRS is tower 7F, 7G, 8F, 8G, ... without the possibility of the IRS3 system!

IES = syngo CT Workplace

The IES console may be installed in parallel to the main operator ICS console via a shared database link. The IES is used as a second console for post-processing applications.



The IES runs only when the ICS is switched on. If the ICS is switched off, the shared database cannot be accessed.

1.4 Software Kit VB42B for ICS, IRS and IES

1.4.1 Software kit VB42B / VB42B_SP1

The software installation kit (p/n **10415490**) comprises the following items required for a complete SW installation. It can also be ordered in **SPC > Service Tools > Software**:

- VB42B software
 - CD_SOM5_IRS_VB42B_V3.0_32bit (part no. 10415484)
 - DVD_SOM5_IRS_VB42B_V3.0_64bit (part no. 10415483)
 - CD_SOM5_ICS/IES_VB42B_32bit_BARE (part no. 10415491)
 - CD_SOM5_IES_VB42B_64bit_BARE (part no. 10415492)
 - DVD_Som5_ICS/IES_VB42B_1of2_32bit (part no. 10415485)
 - DVD_Som5_ICS/IES_VB42B_2of2_32bit (part no. 10415486)
 - DVD_Som5_IES_VB42B_1of2_64bit (part no. 10415487)
 - DVD_Som5_IES_VB42B_2of2_64bit (part no. 10415488)
 - DVD_Recovery, 4 pcs (part no. 10357150)
 - CD Expert-i (part no. 10357782)
 - CD-R 700MB WHITE, 2 pcs (part no. 03078631)



CD_SOM5_IES_VB42B_64bit_BARE & CD_SOM5_ICS/IES_VB42B_32bit_BARE

are the startup CDs used for recovery DVD installation, creation, and integrity checking.

If the system is equipped with IES Tower 8D/E, 10D/E, 11D/E, ***CD_SOM5_IES_VB42B_64bit_BARE*** is needed.

For IES Tower 8 or lower (e.g. Tower 7, 7*) and all ICS Towers:

CD_SOM5_ICS/IES_VB42B_32bit_BARE is needed.



Depending on further developments, the software kit may contain one or more patches (function or service patches). The patches installed in the system may be included on the backup/restore CD. You can also find the patches in the folder C:\Somaris\Service\html\ReportsServicePacks\.

Software installation kit (p/n **11062577**) comprises the following items required for a complete SW installation with computer tower 12 (both ICS and IRS should be computer tower 12). It can also be ordered under **SPC > Service Tools > Software**:

- VB42B_SP1 software
 - DVD_Som5_ICS/IES_VB42B_SP1_1of 2_32bit (p/n 11062566)
 - DVD_Som5_ICS/IES_VB42B_SP1_2of 2_32bit (p/n 11062567)
 - DVD_Som5_IES_VB42B_SP1_1of 2_64bit (p/n 11062568)
 - DVD_Som5_IES_VB42B_SP1_2 of 2_64bit (p/n 11062569)
 - CD_SOM5_ICS/IES_VB42B_SP1_BARE_32bit (p/n 11062570)

- CD_SOM5_IES_VB42B_SP1_BARE_64bit (p/n 11062571)
- DVD_SOM5_IRS_VB42B_SP1_64bit (p/n 11062638)
- CD_SOM5_IRS_VB42B_V3.0_32bit (p/n 10415484)
- DVD_Recovery, 4 pcs (part no.10357150)
- syngo CT Expert-i VA20 (part no. 10357782)
- CD-R 700MB WHITE, 2 pcs (part no. 03078631)



VB42B_SP1 supports the mixed use of towers in ICS, IRS, and IES. For example, the ICS could be T12S, while the IRS may be either T8F or T10F, etc. In addition, the IES could also be T8D/E or T10D/E. If the system does use mixed towers, make sure to take into account the different Windows platforms (32bit/64bit) that will be required.

Tower 12S should have been installed with board D31 which is used in towers prior to ICS tower 11.



In the following context, both VB42B and VB42B_SP1 are referred to as VB42B except specified differently.

1.5 Images and Raw Data



For safety reasons, the customer needs to archive all images and raw data prior to installing the SW, if possible.

1.6 Required System Data



Make a latest backup CD of the system, if possible. This ensures that you have the latest system configuration.

When a new IRS computer is used and installing the IRS software meets with problems, you have to check the computer's BIOS version or update the BIOS first.

For installing the software, the following data should always be available:

1.6.1 Service Key and Serial Number

- From the syngo user interface, select **Options > Local Service**.
- Write down the service key.

	Service key = 20 characters	14 characters	last 6 characters
ICS	Service key (level=7)		
IES 1	Service key (level=7)		
IES 2	Service key (level=7)		

- Write down the serial number.

Computer	Serial Number
ICS	
IES1	
IES2	

1.6.2 Backup Information



When VB42B is newly installed and there is only a previous backup CD on site (not from the VB42B backup), then the directory C:\Somaris\Tools\msi\description must be copied to C:\temp\Tools\msi\description manually before the restore steps are performed. Otherwise, the restore procedures will fail. In addition, the TempCal (delivered with the CT system, if applicable) should be installed after the restore procedures.

To ensure complete backup information, confirm the following:

- No restart is pending due to changes made in the configuration. Open **Local Service > Configuration**, check whether "restart pending" is shown in the message bar. If this is the case, click **HOME** to start the missing automatic restart.
- All applications (e.g. Explorer, Patient Browser) must be closed and stay closed during the entire process of the backup.

1.6.2.1 CD Write Check

During previous installations, we've experienced problems writing the backup files to CD. As a result, this check has to be executed before creating the backup CD.

- Open the local service home menu and go to **Configuration > Offline Devices**.
- Check the offline device.
 - » Depending on the offline device installed, in the drop-down menu select either **CD-R** or **CD/DVD Combo** as an offline device.
 - » Check the item **Default device**, if it is already set to **half speed**, then click **Home** and go to next step. Or select **Half Speed** in the **Recording Speed** item, click **Save**, and then click **Home** to restart.

1.6.2.2 Creating Backup Files

If possible, you should generate new backup files before installing the software.

- Start the service software and select the **Backup & Restore** function in the main menu.
- Insert the backup CD-R.
- Under **Command**, select **Backup**.
- Under **Drives**, select the CD drive, e.g., **[-R-] CD-R** or **[-R-] DVD-R** or **[-R-] DVD**.
- Under **Packages**, select **Software-Setting02-xxx**, **CT-Setting-xxx**, **Security-Settings** and **InSpace** (if applicable) or **SRView-Settings** (if applicable) respectively and click **Go**.
 - » BackupRestore running... Wait until Ready is displayed.
 - » Different backup files will be created.
- Remove the backup medium (**Transfer > Eject from DVD-R**).



Vsim application is removed from VB42B, it is not necessary to backup the Vsim configuration.

1.6.3 Configuration in Service

1.6.3.1 BHC Adaptation (only for ICS of Emotion 6/16 and Emotion 2007)

- From the service software home menu, select **Utilities**.
- Under **Source**, select **Escape to OS** and at the **command** select **NT command Interpreter**.
- Under **Parameters**, type **"C:\Somaris\Tools\Bin\BHC_Configure.exe"**.
- Click **Ok** or press **Return** to activate.
- Record the option checked in the table and click **Close** to close the window.

- Click **Home** to return to the service home menu.

80KV	
110KV	
130KV	

1.6.3.2 Technical Configuration (for ICS only)

- (Continued from the above procedure) Click **Configuration > Technical**, then click **>**, or open the service software home menu and click **Configuration > Technical**, click **>**.



In Emotion 2003, the technical configuration is on one page, it is not required to click **>**.

- Under the **Quality Configuration** item or **Technical Configuration**, record the value of **Language for Quality PDF Protocols** and **Country for Quality**.

Language for quality PDF protocols	
Country for Quality	

- Under the **Constancy** item or **Quality StartUp Message**, record the checkbox status of each item (if available), e.g.: Calibration, Phantom Check...

Included:	Expired:	out of tolerance	Included:	Expired:	out of tolerance
Calibration			MTF		
Phantom check			Contrast		
Lightmarker			Low contrast		
Sag./Cor.Lightm			Table position		
Prev.Image pos.			CTDI Air		
Slice			CTDIw Head		
Homogeneity/Water			CTDIw Body		
Noise					

1.6.3.3 SRS Status Check

- (Continued from above procedure) Click **SRS** under **Service**, or from the service software home menu, select **Configuration > SRS**.

- Click > to start the SRS function check.



The function name may be different depending on the software running on the system. Virus protection is always disabled in VB42B.

Note the result of the SRS function check (SRS status) in the following table:

Contents	ICS		IES1		IES2	
SRS functions	Desired	Actual	Desired	Actual	Desired	Actual
SRS Basic Function (*)						
HTTPS Proxy (Netviewer, CSKB)(*)						
Utilization Management (**)						
System Management (**)						
Virus Protection (**)						

(*).... mandatory SRS functions, predefined within Somaris

(**). optional, the checkbox is enabled only if the customer purchased this EPS service (EPS=Enhanced Productivity Service)

- If you have changed something, you have to click **Save** and then click **Home**. If is re-boot is required, confirm it. If not, close the service window.

1.6.4 Configuring in Application

1.6.4.1 Regional Settings

- From the syngo user interface, select **Options > Configuration**.
- In the configuration panel, double-click the **Regional and Language Options** icon.

Write down the following data:

Standards and format		Customized	Time > Time For- mat		Time separa- tor	
			Date > Date for- mat		Date separa- tor	
Location						
Language	Details		Input language			
	Used in menus and dialogs					

1.6.4.2 Time Zone Configuration

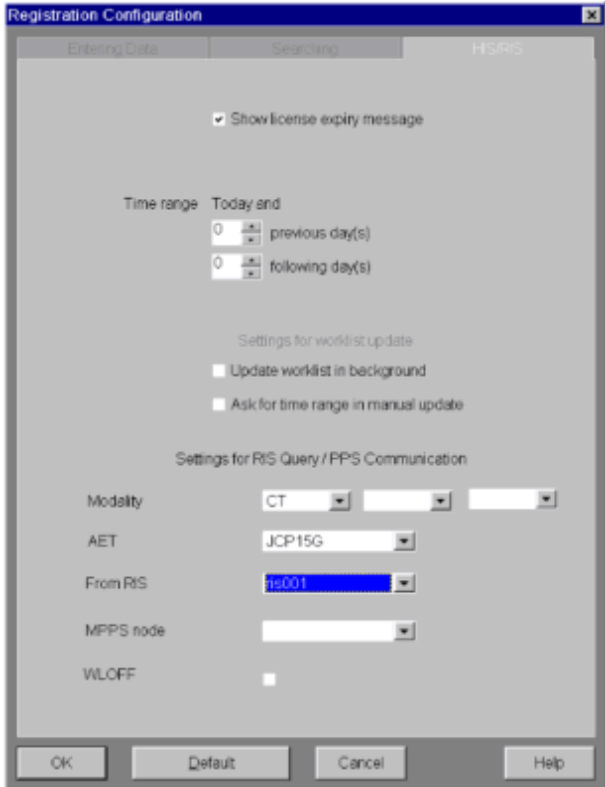
- (Continued from the above procedure) Double-click the **Date and Time** icon, or from the syngo user interface, select **Options > Configuration**. and double-click the **Date and Time** icon.
- Check the time and record the time zone in the table

Time Zone	
-----------	--

1.6.4.3 HIS/RIS Configuration Check (optional)

- From the service software home menu, select **Options > Configuration**.
- In the configuration panel, double-click the **Patient Registration** icon.
- Under the **HIS/RIS** tab card, record the value/status of: **Modality, AET, From RIS, MPPS node, WLOFF**.

Fig. 1: Only an example



Modality		
AET		
From HIS		
MPPS node		
WLOFF		

1.6.4.4 Remote Service Access Configuration

- In the syngo user menu, select **Option > Service > Remote Service**.
- Record the remote access mode in the table.

Limited Access Permanent		
--------------------------	--	--

Limited Access		
No Access		

1.6.4.5 Report.htm and License.dat

In case a backup file becomes corrupted, it is useful to have both the system configuration and the license file available. For this reason, you have to save these files.

- Open Windows Explorer.
- Copy the **report.htm** file in the **C:\Somaris\service\html\report** folder to, e.g., a storage place.
- Copy the **license.dat** file in the **C:\Somaris\Config\Licensing** folder to, e.g., a storage place.

1.6.4.6 User-Specific Data



To prevent missing data, the CSE needs to manually store the postprocessing applications data shown below (which is on the H: drive) for the customer prior to installing the update.

These data have two use cases: **Report creation for a normal clinical workflow** and **extractions of data for special use cases such as clinical investigations, papers, presentations**. The following data (if available, depending on system configuration) in the **H:\Sitedata** folder need to be stored manually.

- CaScoring
- DynEva
- InSpace
- Lungcare
- Pulmo
- syngoCirculation
- syngoColon
- Quality Reports (ICS only)

If you have a storage medium, you have two ways to make the backup:

- Select **Options > File Browser**.
- Select the folders and send them to the storage medium.

or

- Open an Explorer window and copy the folders to the storage medium.

1.6.5 Additional Steps for the Sliding Gantry

Backup of system tables for the sliding gantry:

- Open **Windows Explorer**.

- Open the **C:\Somaris\config\ControlSystemTables** directory
- Select the **T3A, T3C, T36, T17, T37, T1A, T1B, T30** files (if available), **T3F** files (if available) and copy these files to a storage medium.
- Change to the **C:\Somaris\config\ControlSystemFlash** directory.
- Select the **version.txt** file and **PHSCV****.FSH** (if available), and copy to a storage medium.
- Change to the **C:\Somaris\config** directory.
- Copy sub-directory **P15PHS** (if available) and **PHS1** (if available) to a storage medium
- Open the **C:\Somaris\tools\msi\descriptions** directory.
- Copy all files and all sub-directories to a storage medium.

Note Scannable_Range:

- Open the '**C:\Somaris\config\s_SOMARIS_General.spec**' file with Notepad to view the following two lines in the file:
- Note the values for the scannable range:
 - Write down the value for 'SCANNABLE_RANGE : [0 160]'
 - Write down the value for 'SCANNABLE_RANGE = 160'

SCANNABLE_RANGE:e.g.[0 160]	
SCANNABLE_RANGE=e.g. 160	

(Note: 160 is only an example; please write down the correct values for your systems)

1.7 Things to remember:



During the software installation of the ICS, the IES has to be switched off.



Unplug the network cable between the ICS and the Ethernet switch (disconnect the external network). After the software installation, plug in the cable again.



Walk through the installation step by step and do not try to save time by working out your own "installation strategy". This is especially important when installing the IES. WAIT when it asks you to wait until the ICS is ready.



Ensure that the Patient Browser is never opened during the installation.

2.1 Software Installation



A total of 2 installation methods can be used. One is using the Recovery DVDs, the other is using the delivered VB42B DVDs.

It is recommended to use the Recovery DVDs for installation.

Only if the Recovery DVDs are not functional, should you use the delivered VB42B software DVDs for the installation. After the installation, please create new Recovery DVDs.



Refer to Chapter 1 for configuration preparations (→ Required System Data / Page 12).



In the following context, installation CD/DVD of VB42B is mentioned. When VB42B_SP1 is installed, the corresponding VB42B_SP1 CD/DVD should be used instead.

2.1.1 Starting the Installation using Recovery DVDs

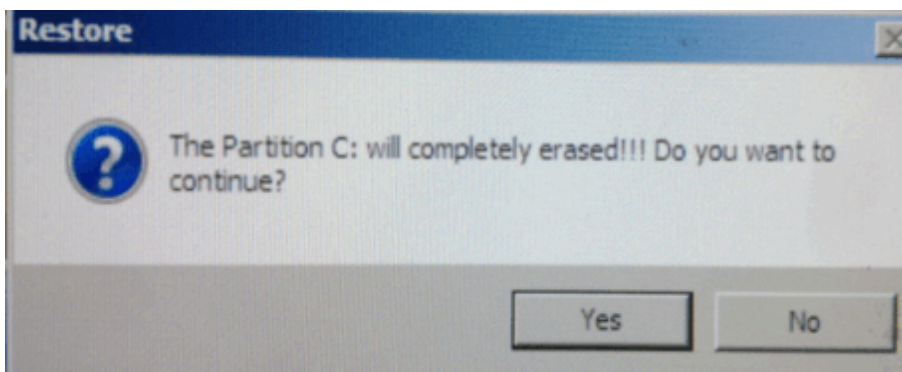
- Insert the **CD_SOM/5_ICS/IES_VB42B_32bit_BARE** at the ICS
- Select **System> End> Restart system** and boot from the CD.
- Disconnect the external Ethernet cable of the ICS (normally connected to the additional network card).



If the external network is connected, you may experience an IP address conflict

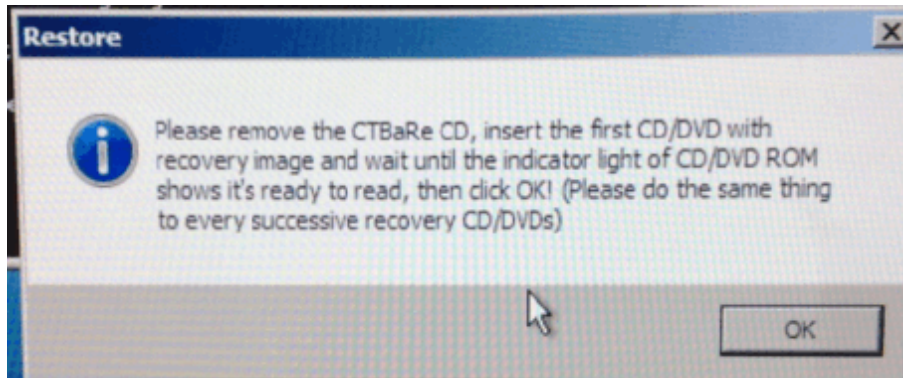
- After the system reboots successfully, select **Restore the System Drive**, and press **OK**.
 - » The following message appears, press **Yes**.

Fig. 2:



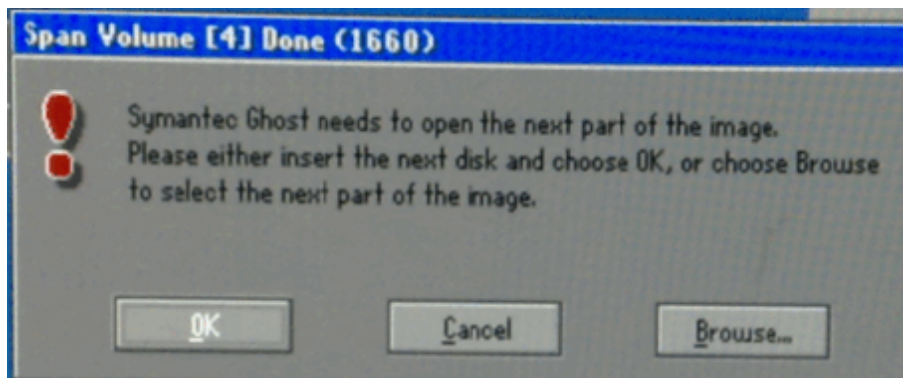
- The following message appears: Please remove the CTBaRe CD, insert the first CD/DVD with recovery image,

Fig. 3:



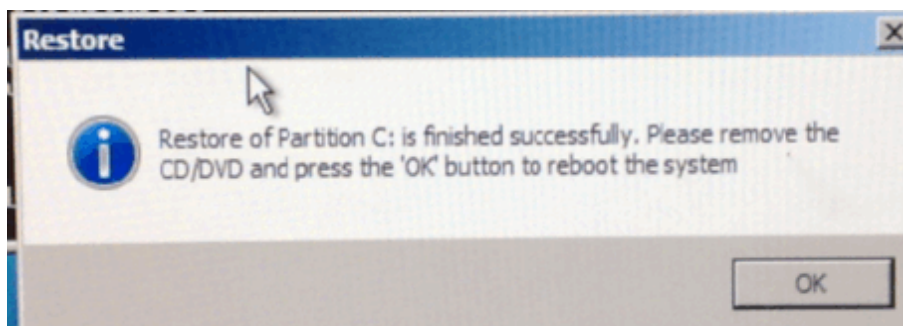
- Insert the first Recovery DVD. When the DVD is ready, press **OK**.
- The system drive is being restored. This will take about 15 min.
- Insert the 2nd DVD when you are asked. Click **OK** when the DVD is ready.

Fig. 4:



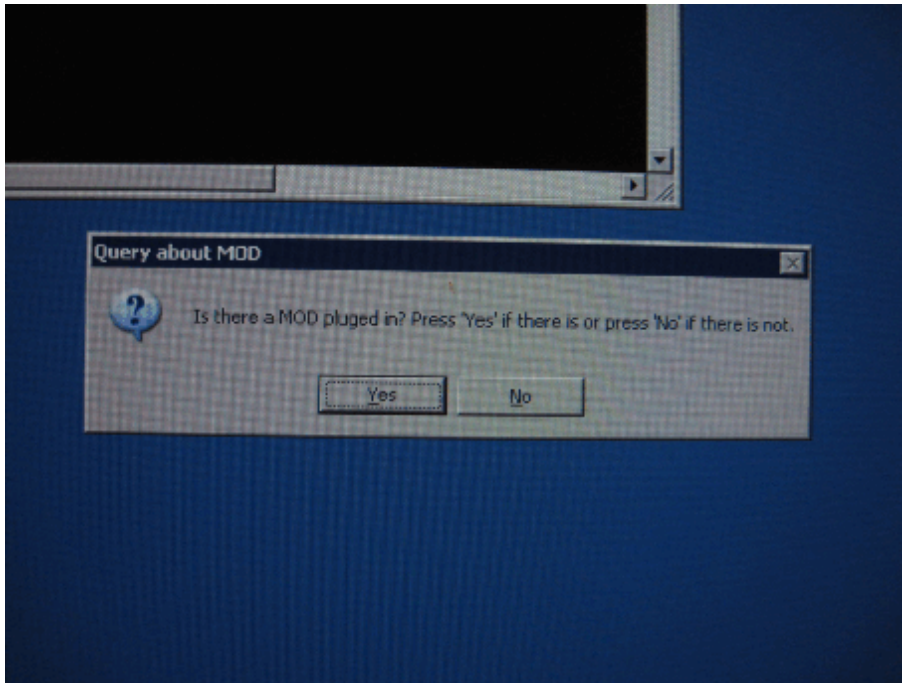
- When the restore process is finished, the message appears: Restore of Partition C: is finished successfully... Remove the DVD and click **OK** to reboot the system.

Fig. 5:



- Wait until the message *Is there a MOD plugged in?.....* appears, click **Yes** or **No** according to whether you have one. If the ICS computer does not have a MOD, you should click **No**.

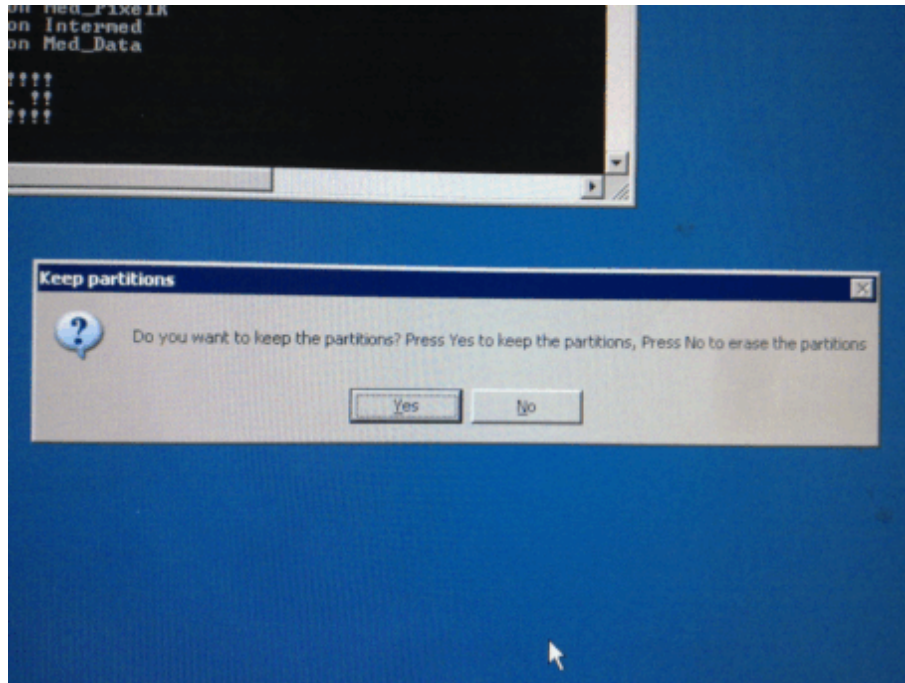
Fig. 6:



- » The system will check the partitions.

- A window prompts with the message "Do you want to keep the partitions? Press Yes to keep the partitions, Press No to erase the partitions. Press **Yes** or **No** to proceed.

Fig. 7:



Only respond with Yes if the same software version is installed and the data disks do not have to be replaced. When No is pressed, all patient data will be lost.

- Wait for about 5 min., the system will reboot automatically.
- After the system has been successfully started, select **Option > Service > Local Service** to enter into the service software and Go to the Restoring Steps (→ Restoring Steps / Page 29).

2.1.2 Starting the Installation using the delivered VB42B DVDs

- At the IES (if it exists), shut down all applications and switch off the IES system.
- Insert the **DVD_Som/5_ICS/IES_VB42B_1_of_2_32bit** in the ICS.
- At the ICS, select **System > End** and **Shut down** the system.
- Switch off the CT system.
 - For an Emotion system, switch it off at the UPS.
 - For a Sensation system, switch it off at the control box.
- Disconnect the external network cable of the ICS.



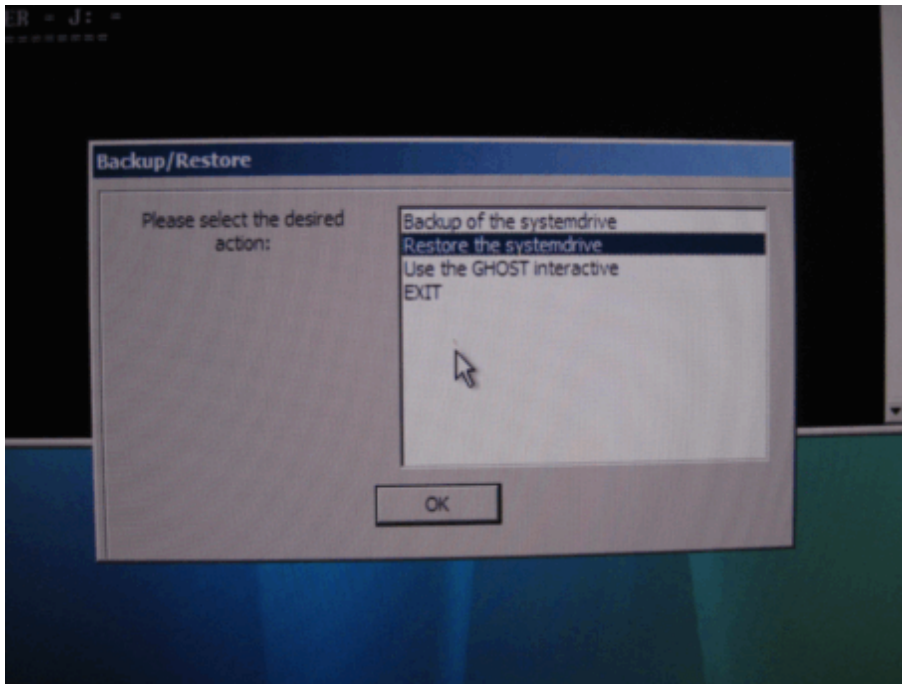
If the external network is connected, you may experience an IP address conflict

- Switch on the system at the UPS (computer "on").

The computer boots from the CD/DVD drive. After a short while, the message `Windows is loading files...` appears. Wait until a Backup/Restore window shows up.

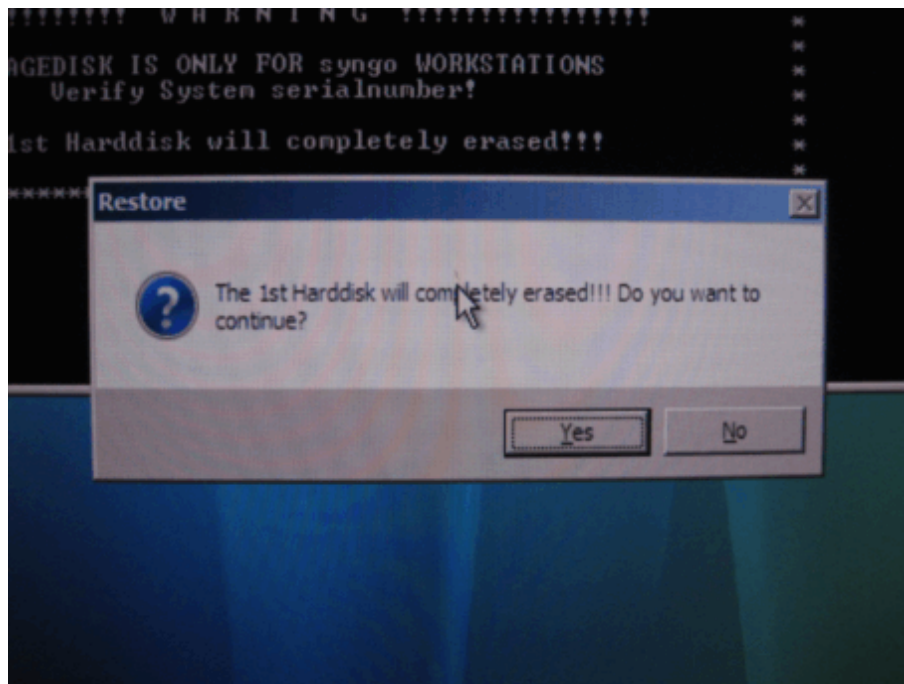
- » Select **Restore the systemdrive** item using the mouse or (if the mouse does not respond) the Tab key and arrow keys, then click **OK**.

Fig. 8:



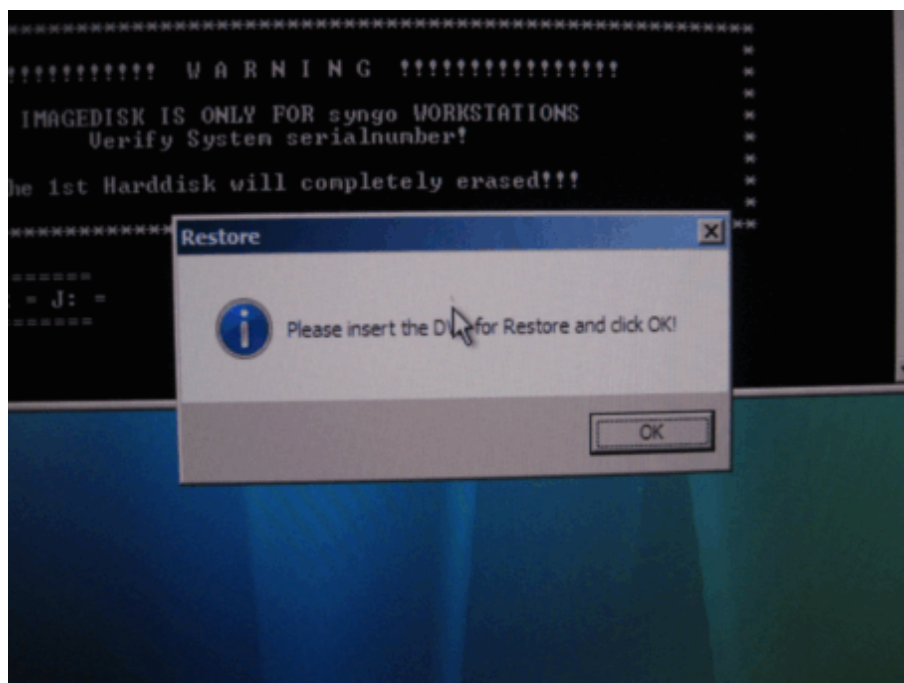
- A message "The 1st Harddisk will be completely erased!!! Do you want to continue? pops up. Click **Yes**.

Fig. 9:



- A message Please insert the DVD for Restore and click OK pops up. Click **OK**.

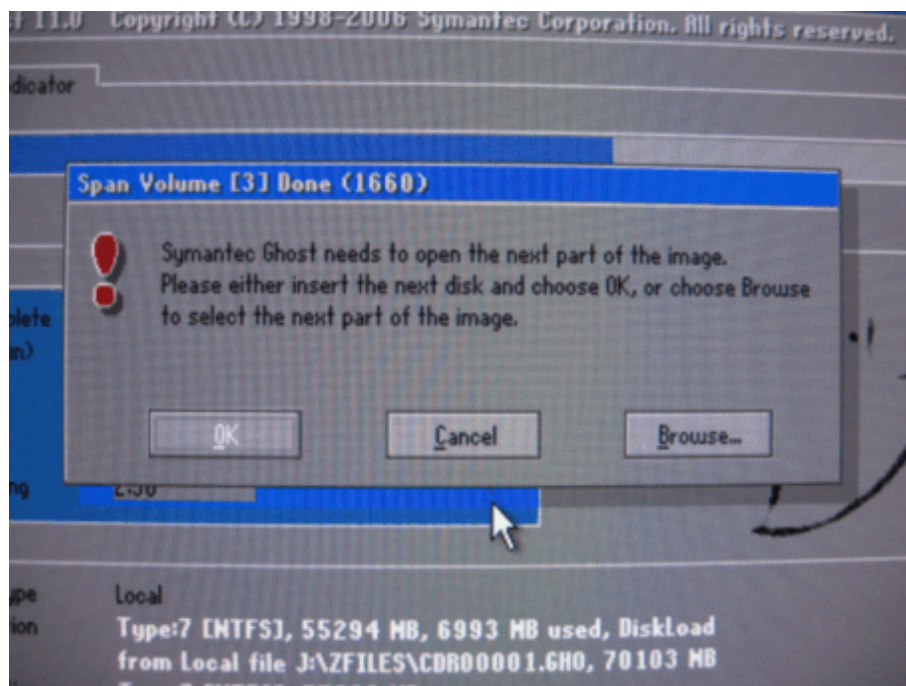
Fig. 10:



- » After the disk drives check, the GHOST is running. The system is copying files.

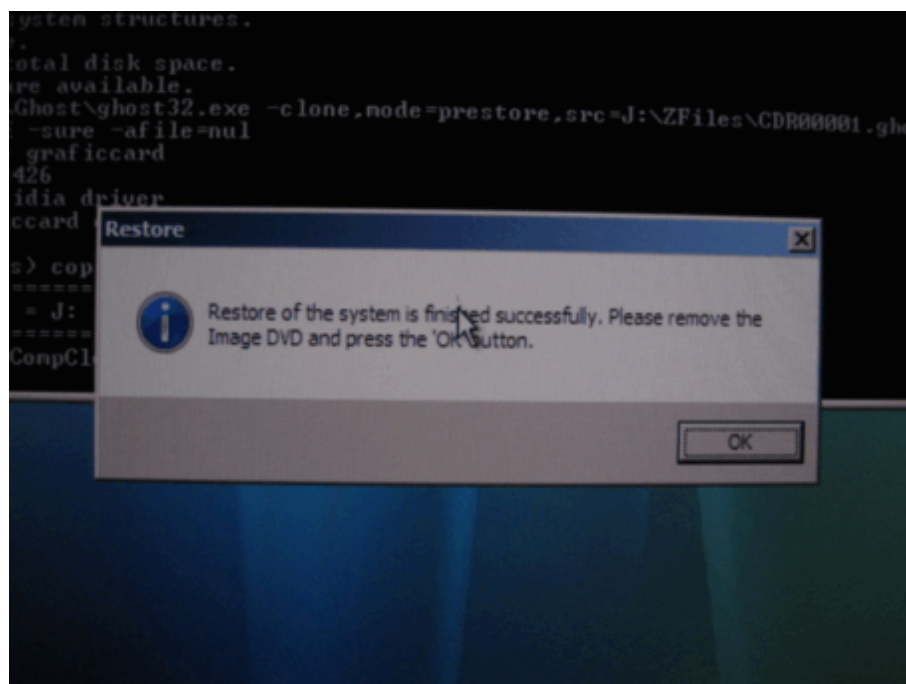
- Insert the **DVD_Som/5_ICS/IES_VB42B_2_of_2_32bit** when the system asks for it, and click **OK** to confirm that the DVD is ready.

Fig. 11:



- Wait until the DVD ejects automatically. At the same time, the message **Restore of the system is finished successfully...** appears. Remove the DVD and click **OK**.

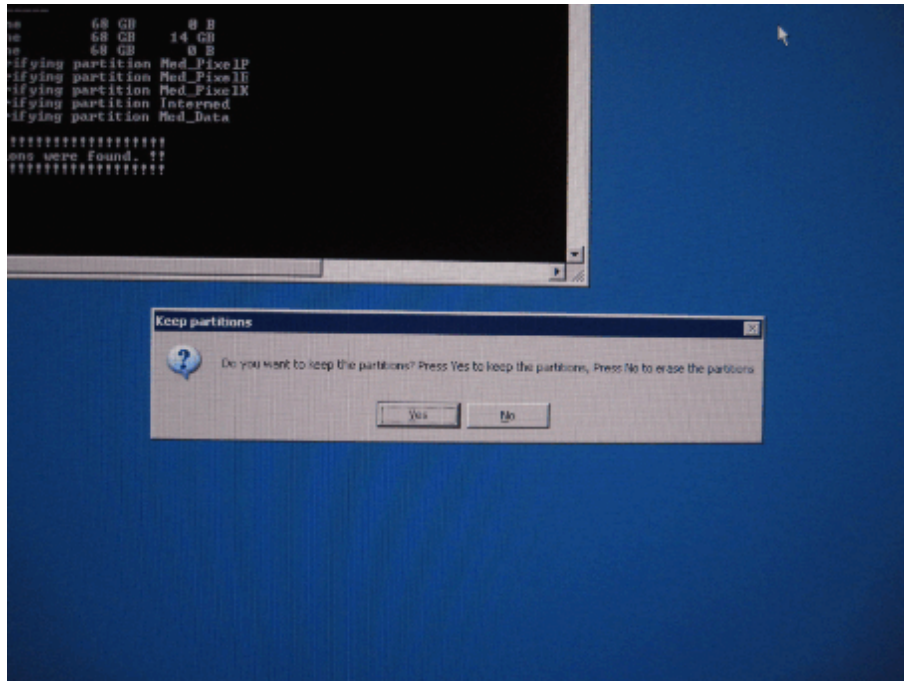
Fig. 12:



- The system restarts automatically and continues with the installation.
- Reboot will take place.

- Wait until the message Do you want to keep the partitions? ...appears, click **No** to erase all the partitions. Or click **Yes** to keep all the data partitions.

Fig. 13:



Only select Yes if the system previously had the same software version, and the data disks must not be replaced.

- Installation is running and rebooting will take place.
- Wait until the following screen appears (the picture is only an example).

Fig. 14: (This is only an example)



- Select the correct Product Name to be installed.

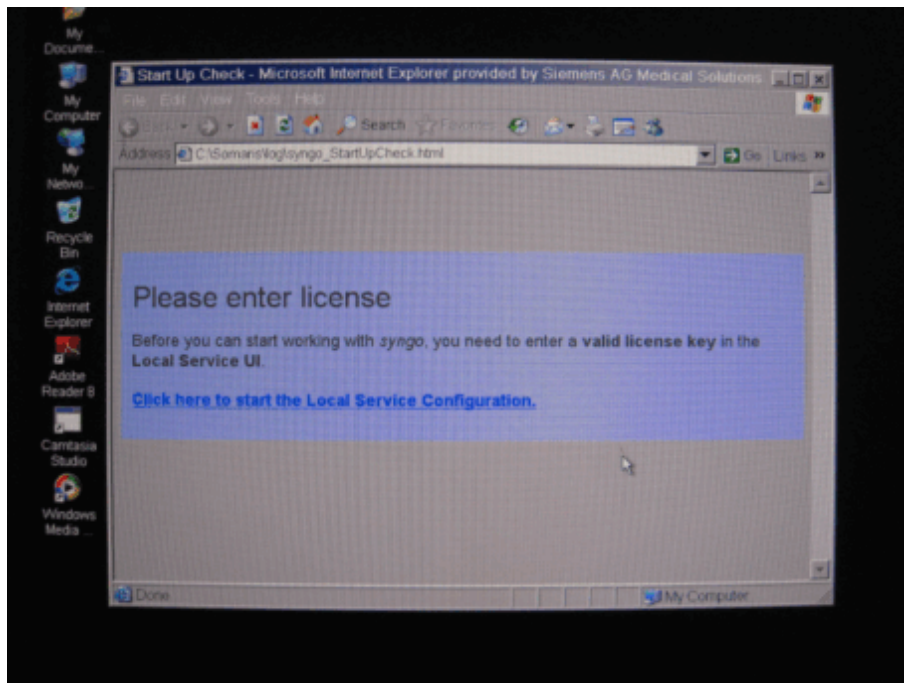


For selection of the product name, please refer to SPC or CB-DOC. Select the product name according to the serial bands in the SPC or CB-DOC > function description.

- Click **Start**.

Completion of the ICS installation (approx. 25 minutes) is indicated by a pop-up window with the message **Please enter license**.

Fig. 15:



- Click **Click here to start the Local Service Configuration** to start the service software (complete service key and system serial number are required).

2.2 Restoring Steps



Restoring steps have to be implemented after the software installation, using the Recovery DVDs or original software DVDs.

Before starting the restore steps, make sure the IRS is ready and connected to the ICS.



When VB42B is newly installed and there is only a previous backup CD on site (not from the VB42B backup), then directory C:\Somaris\Tools\msi\description must be copied to C:\temp\Tools\msi\description manually before the restore steps are performed. Otherwise, the restore procedures will fail. In addition, the TempCal (delivered with the CT system, if applicable) should be installed after the restore procedures.

- Open windows **Start** menu, click **Run** and enter **notepad** ↵.
- Exactly enter the following content in the notepad.exe program:

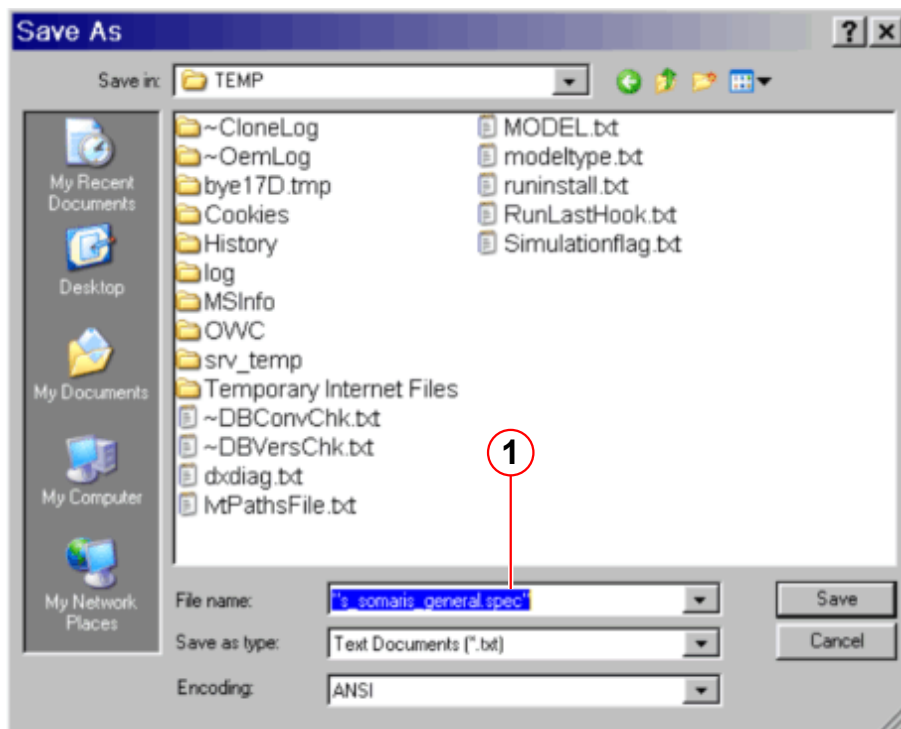
```
PHS:int
PHS=0
#end
```
- Save the file as "C:\temp\s_somaris_general.spec" as shown in the picture and close the notepad.exe program.



It is important to have quotation marks at both ends of the string.

The procedure is the same as for "s_somaris_general.spec".

Fig. 16: Saving file



(1) File name and quotation marks

- In the windows **Start > Run**, enter **cmd** ↵, and in the command window, enter **CD C:\temp** ↵, followed by **configconvert C:\temp\somaris_general.spec -u** ↵.

Fig. 17: Command of configconvert



- Enter into service software, go to **Utilities > Escape to OS**. Under **Command**, select **NT Command (secure)**, at **Parameters**, type **type nul > C:\Somaris\config\ControlSystemTables\IP15PHS** ↵ (activate with return at the end of the command)
 - » There may be different results in different systems. Neglect whatever the result is.
- Open Windows Explorer, and check if there is a file or folder **P15PHS** under the **C:\Somaris\config\ControlSystemTables** directory.



The above steps are the preparations for restoring the PHS settings from the backup CD. If an error message pops up, neglect it, such as access denied. After you have completed the above, there should be a folder or a file named P15PHS under the C:\Somaris\config\ControlSystemTables directory. Otherwise, the restoring step will be blocked and the system won't come up.

2.2.1 Restore 1

- In the main menu of the service software, select the **Backup & Restore** function.
- Insert the ICS backup CD.
- Under **Command**, select **Restore**.
- Under **Drives**, select the corresponding drive.
- Under **Archive**, select **SW-Settings02-XXX**.
- Under **Groups**, select all items (select the first item, then with **Shift** key pressed, select the last item) and click **Go**.
 - » BackupRestore Running ...; wait until Ready is displayed.



Ignore the error ".....C:\Somaris\log\FilmRestore.log for more detailsUpdate.bat finished-File not found - Layout".



Ignore the error ".....The host name of the current machine is different from the host name in the security settings backup."

- Under **Archive**, select **CT-Settings-xxx**.
- Under **Groups**, select all items except **IRS** and click **Go**.
 - » BackupRestore Running ...; wait until Ready is displayed.

- Under **Archive**, select **SecuritySettings-xxx**
- Under **Groups**, select **all** and click **Go**.
 - » `BackupRestore Running . . .`; wait until `Ready` is displayed.



Ignore the error ".....The host name of the current machine is different from the host name in the security settings backup."

- Click **Home**, and confirm the system reboot.
- Connect the external network cable of the ICS.
- After the system reboots successfully, confirm all prompted windows.



The prompted window may be a note telling of a Firmware update, or SolidCore Startup Error, Quality Hint, Checkup. Cancel the checkup at this time.

2.2.2 Restore 2



Make sure the IRS is up and ready, the LAN cable between the IRS and ICS is connected.

- In the main menu of the service software, select the **Backup & Restore** function.
- Under **Command**, select **Restore**.
- Under **Archive**, select **CT-Settings-xxx**.
- Under **Groups**, select **IRS** and click **Go**.
 - » `BackupRestore Running . . .`; wait until `Ready` is displayed.



Only for Sensation systems:

Click **OK** in the message window regarding **Cupping Correction** and execute it when completing the configuration(→ Cupping Correction (only for Sensation system) / Page 34).

- In the syngo user menu, eject the backup CD by clicking **Eject from DVD-R** under the **Transfer** menu. You have to wait a while for the ejection.
- In the service software window, click **Home**, confirm the system restart. At this time, you have to ensure that the gantry is powered on if it was off before.



After the system restarts successfully, you may go on with the following steps while the system is updating the firmware.

2.2.3 Restoring the Dynamic Backup File

If the dynamic backup file is newer than the CT-Settings, perform the restore of the dynamic backup files. Otherwise, skip this section.

- Under **Drives**, select the drive which contains the dynamic backup package.
- Under **Archive**, select **..._BackupDynamicData_...**
- Under **Groups**, select **AllDynamicData** and click **Go**.
 - » `BackupRestore Running ...`; wait until **Ready** is displayed.



Neglect the error(s) encountered.

- Click **Home** and accept the automated reboot with **OK**.

2.2.4 Additional Steps for the Sliding Gantry (option, only with sliding gantry)



Before performing the following procedures, you have to enter the configuration menu of the service software and enable the sliding gantry mode.

This work step will be performed only if there is a sliding gantry installed on-site:

- Insert the CD firmware (delivered with the system) "**Sliding Gantry P10 Emotion/Emotion 2003**" or "**CD_Sliding Gantry_Emotion 2010**"(p/n 10415243) into the ICS drive.
- In the service main menu, click **Utilities**, select **Escape to OS, NT Command Interpreter**, under **Parameters**, type in `sadmin bu` and activate with **Enter** or **GO**.
- Open Windows Explorer by typing in `Explorer` under **Parameters**.
- Select the ICS drive and go to the **R:\Emotion 2003** or **R:** directory.
- Double-click "**install_sliding_copy.bat**".
- Change to the **C:\Somaris\Config\SlidingGantry** directory, double-click "**install_sliding_2.bat**". Continue the installation according to the instructions on-screen (by pressing any button twice).
 - » The tables are copied into **C:\Somaris\Config\ControlSystemTables**.
- Insert the backup medium into the computer.
- Select **T3A, T3C, T36, T17, T37, T1A, T1B, T30** files (if available), **T3F** files (if available) in the medium.
- Copy back to the **C:\Somaris\config\ControlSystemTables** directory to overwrite the originally installed ones.
- Select the **PHSCV****.FSH** file (if available) in the medium and copy it back to the **C:\Somaris\config\ControlSystemFlash** directory
- If the previous step has been completed, edit the **version.txt** file to have **PHSCV****.FSH** replace the original **PHS*****.Fsh** in the file.
- Change to the upper layer directory and copy back the **P15PHS** directory (if available) and **PHS1** (if available) from the storage medium.

- Change to the **C:\Somaris\Tools\msi\description** directory.
- Copy back and overwrite the **description_files** directory, **P15PHS** (if available) and **PHS1** (if available) from the storage medium.

Check Scannable_Range:

- Open the '**C:\Somaris\config\s_SOMARIS_General.spec**' file with Notepad to view the following two lines in the file:
 - SCANNABLE_RANGE :
 - SCANNABLE_RANGE =
- Modify the values for the scannable range according to the record:
- In the service window, type in `sadmin eu` in the **Parameters** field and activate with **Enter** or **GO**.
- Close the service window.

2.2.5 Configuration Walk-Through

- In the main menu of the service software, select **Configuration**.
- Click **Next** on the **Initial setup** page
- Set the Administrator password to the default string and click **Save**.



This password should be set to the default string, otherwise, the "SolidCore startup failed" window will show up when the system starts.

- Click **Next** or **>** for the pages in the later pages, until both **Finish** and **Save** buttons appear on the last page.



The MIP rendering algorithm under "Application and >" has to be set manually, since it depends on the system's hardware.

- Click **Save** and then **Finish** to save the settings.
- Delete the Vsim license if it is in the license file:
 - Open Windows Explorer and open the **C:\Somaris\Config\Licensing\license.dat** file with NotePad or WritePad.
 - Search Vsim and delete the whole license line (from "Feature VSIM or xxx_VSIM" to the beginning of the next feature) if applicable.
- Click **Home** and confirm and wait for application restart, if applicable.

2.3 Completion of Configuration

2.3.1 BHC Adaptation (only for ICS of Emotion 6/16 and Emotion 2007)

- From the service software home menu, select **Utilities**.
- Under **Source**, select **Escape to OS** and under **Command**, select **NT Command Interpreter**.
- Under **Parameters**, type in **C:\Somaris\Tools\Bin\BHC_Configure.exe**.
- Click **Ok** or press **Return** to activate.
- Set the options according to the previous record, and click **Start**.
- Click **Close** to close the window, and click **Home**.

2.3.2 Tuning CD Burning

- In the service software menu, select **Utilities**.
- Under **Source**, select **Tune CD Burning**.
- Activate the **Tune CD Burning for current hardware** checkbox.
- Activate with **Return** or **Go**.
- Confirm the prompted dialogue. Wait until the `Drive has been successfully installed` message appears and click **OK**.
- Click **Home**.
- Close the service window.

2.3.3 Cupping Correction (only for Sensation system)

- From the Service Home Menu, select **Utilities > Escape to OS**
- Navigate to **C:\Somaris\tool\bin**.
- Execute **Configure_CupCor.exe** (double-click file with mouse)
- Do not modify settings in the upcoming window and click **Start**

2.3.4 Remote Service Access

- In the syngo user menu, select **Options > Service > Remote Service**.
- Check and set the option according to the previous record.
- Click **OK** to close the window.

2.3.5 Copying back user-specific Data

Open Windows Explorer and copy back the user-specific data from the storage medium to the **H:\Sitedata** directory. When asked for “confirm folder replace”, click the **Yes to All** button.

2.3.6 Importing Reference Images (optional)

- From the Patient Browser, select **Transfer > Import from Offline**.
- From the drop-down list, select the **H:\SiteData\ReferenceImages** path.
- Select all images and click **OK**.
 - » After the import, the images are available in the Patient Browser.

2.3.7 Setting the Time and Time Zone

- In the syngo user menu, select **Options > Configuration**.
- Double-click the **Date and Time** icon, set the correct date, time and time zone.



Do not activate the NTP time server or select automatic Daylight Savings Time.

- Click **Apply** and then **OK** to close the properties' window.

2.3.8 Configuring the Keyboard Language

Perform the following steps to configure keyboard languages other than English (**except for Chinese and Japanese**):

- From the syngo user interface, select **Options > Configuration**.
- In the configuration panel, double-click the **Regional and Language Options** icon.
- Set the proper language and location.
- Click **Customize...** and set the time and date format according to the previous record and click **OK**.
- Select the **Languages** tab card, set the language to be used in the menu and dialogue windows and then click **Details**.
- Click **Add**, select the correct input language and confirm with **OK**.
- Select the English keyboard language, click **Remove**, and confirm with **OK**.
- Click **OK** to close the window.
- Click **OK** to confirm the change.
- Confirm restart of the system.

Attention: For **Chinese/Japanese:** Open the **DICOM/Character Set** dialog in Local Service. Select the corresponding character set, for Japanese "**Japanese, Katakana**" and check the option **Allow character set extension**; for Chinese **Unicode** or **Chinese**. Click **Save** and click **OK** to confirm the saving. Click **Home** and confirm the application restart.



If a blank web page appears for some time, the computer has to be restarted manually.

2.3.9 HIS/RIS Configuration Check (optional)

- From the syngo user menu, select **Options > Configuration**
- In the configuration panel, double-click the **Patient Registration** icon.
- Under the **HIS/RIS** item, change the value/status along with the record.

2.3.10 Setting the Workflow

If the customer requires Auto Filming, he can select this under **Options > Configuration > Examination**.

2.3.11 Additional Steps for the ICS and IES Interconnection



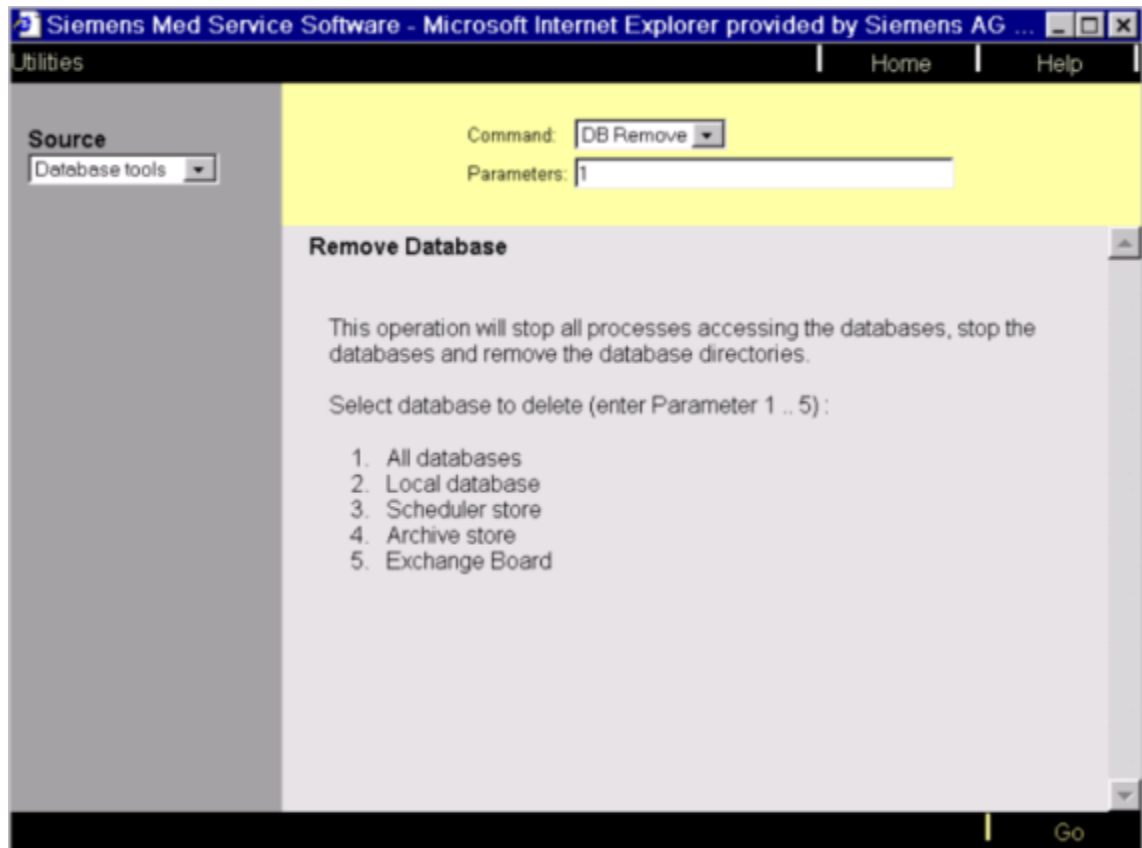
If a system has an IES connected, a message may display in the IES interface "**Database creation of exchangeboard failed - please call service**".

Please execute the following instructions on the IES to resolve this problem.

- From the service software home menu in IES, select **Utilities**.
- Under **Source**, select **Database tools** and under **Command** select **DB Remove**

- Under **Parameters**, type in 1.

Fig. 18: DB Remove



- Activate with **Return**.
 - » A message `Are you sure, you want to ...` appears. Click **OK**.
- The system will reboot automatically.

2.4 Known Bugs and Errors

The following points are known issues and need manual intervention.

1. For dual monitors on the ICS or IES, the syngo task bar will be displayed at a higher position than normally. As a result, the relevant parts of the syngo User Interface are not visible. This can be corrected as follows (under the administrator's account):
 - Open **Start->Settings->Control Panel->NVIDIA nView Desktop Manager**.
 - Deactivate the nView Desktop Manager.
 - Restart the system.
 - After restarting the system, the failed task bar should be fixed.
2. The MIP rendering algorithm under "Application and >" has to be set manually, since it depends on the system's hardware.
3. If a computer lower than Tower 11 is replaced with Tower 11 or higher, the API (Automatic Patient Instruction) settings from the lower- version Tower is of worse quality due to the new sound chip. We recommend to record the API sound files again.

3.1 Installation of the IRS

3.1.1 SW Installation on the IRS Tower

Select the correct CD per the following table.

Tower	Software installation CD
IRS - Tower 7A/7B	IRS_VB42B_V3.0_32bit
IRS Tower 7G	IRS_VB42B_V3.0_64bit
IRS Tower 8A/8B	IRS_VB42B_V3.0_32bit
IRS Tower 8F/8G	IRS_VB42B_V3.0_64bit
Tower	IRS_VB42B_V3.0_64bit



In the following context, installation CD/DVD of VB42B is mentioned. When VB42B_SP1 is installed, the corresponding VB42B_SP1 CD/DVD should be used instead.

- Insert the **DVD_SOM5_IRS_VB42B_V3.0_64bit** or **CD_SOM5_IRS_VB42B_V3.0_32bit** into the CD/DVD drive.
- Switch the computer off and on again to reboot.
 - » The computer boots from the CD/DVD, and the CD/DVD drive will flash continuously.
- Wait until the CD/DVD is ejected and stays ejected for more than 5 seconds. Subsequently, take out the CD/DVD.



If there is problem installing it, you have to check the BIOS version or the BIOS has to be flashed with the latest one. As for the latest BIOS version, you can download it from the Intranet: <https://bps-healthcase.siemens.com/productinformation/> CT->Software download->Spare Parts Replacement.

The surface of the CD/DVD should be kept clean.

- Keep pressing the power button in the IRS for more than 5 seconds to power it off. Then press the power button again to power on the IRS.
 - » The computer starts the SW installation and will be finished within approx. 40 minutes.



You can reach the IRS via the Remote Desktop to clarify if the IRS software installation has been finished or not:

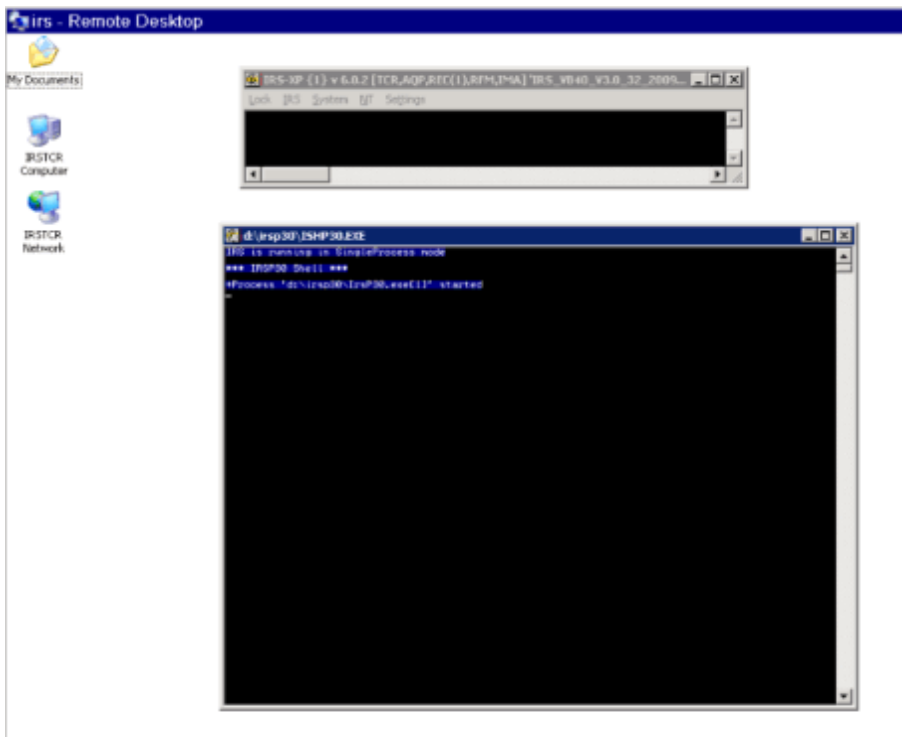
1. At the ICS, click **Start > Programs > Accessories > Remote Desktop Connection**
2. Input the computer name **irs**.
3. In the upcoming windows, input the user name **irs**, and the password **IrsP30..**
4. Clarify if the IRS installation is finished or not.

In the picture below, the IRS installation has been completed and is ready.

Fig. 19: Remote Desktop to IRS



Fig. 20: IRS is ready (only an example)



- If the IRS software installation has been completed, power off the entire computer system, and power it on again.
- Continue with the IRS configuration(→ IRS Configuration / Page 41).

3.1.2 SW Installation for IRSmx2s Computer

After replacing the IRSmx2s, the software has to be installed at the computer.

- Insert the **DVD_SOM5_IRS_VB42B_V3.0_64bit** or **CD_SOM5_IRS_VB42B_V3.0_32bit** into the CD/DVD drive of the replaced IRSmx2s computer.
- Switch the computer off. Wait for 5 seconds and switch the computer back **ON** with the On/Off button at the dashboard.
 - » The computer boots from CD or DVD and the SW installation begins.
- Wait until the CD/DVD drive is open (approx. 10 min.), remove the CD/DVD.
 - » The IRS installation is finished within approx. 30 minutes.
- When the IRS software is finished, continue with the IRS configuration(→ IRS Configuration / Page 41)



If both ICS and IRS are installed at the same time, and the ICS has not completed its restore, the IRS configuration can be skipped.

3.1.3 IRS Configuration

- After the IRS installation is finished, enter the Service Key in the ICS system and open the Local Service main menu.
- From the service software home menu, select **Utilities**.
- Under **Source**, select **Escape to OS** and under **Command**, select **NT Command Interpreter**.
- Under **Parameters**, type in **C:\Somaris\utils\IrsTablesInstall.bat**.
- Activate with **Return** or **Go**.
 - » The default tables are transferred to the IRS.



If the message `...please waiting...` shows up in the window for more than 1 minute, please unplug the ICS & IRS power cable and wait for 3 minutes. Afterward, plug in the power cable to restart the ICS & IRS again.

- When the transfer is completed, click **Home** and select the **Backup & Restore** function.
- Insert the ICS backup CD.
- Under **Command**, select **Restore**.
- Under **Drives**, select the corresponding drive.

- Under **Archive**, select **CT-Settings-XXX**.

Fig. 21: (This is only an example)

Command	Drives	Groups
Restore	[R-] CD-R	Mode
		Cardio
	Archive	Tube
	<hostname>CT-Settings--PXXX_date_time.ar	Service
		IRS
		General
		Firmware

- Under **Groups**, select **IRS** and click **Go**.
 - » The "site-specific tune-up tables" are transferred to the IRS.
- When **Ready** is displayed, remove the ICS backup CD and restart the complete system.

4.1 Software Installation



A total of 2 installation methods can be used. One is using the Recovery DVDs, the other is using the delivered VB42B DVDs.

It is recommended to use the Recovery DVDs for installation.

Only if the Recovery DVDs are not functional, you should use the delivered VB42B DVDs for installation. After the installation, please create new Recovery DVDs.



CD_SOM/5_IES_VB42B_64bit_BARE & CD/SOM/5_ICSI/IES_VB42B_32bit_BARE

are the startup CDs used for recovery DVD installation, creation, and checking.

If the system is equipped with IES Tower 8D/E, 10D/E, 11D/E, ***CD_SOM/5_IES_VB42B_64bit_BARE*** is required.

For IES Tower 8 or lower (e.g. Tower 7 or 7*),

CD_SOM/5_ICSI/IES_VB42B_32bit_BARE is necessary.



Refer to chapter 1 for configuration preparations (→ Required System Data / Page 12).

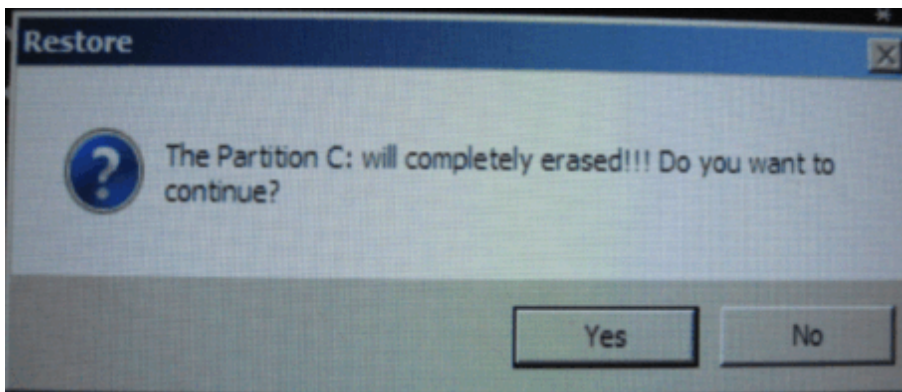


In the following context, installation CD/DVD of VB42B is mentioned. When VB42B_SP1 is installed, the corresponding VB42B_SP1 CD/DVD should be used instead.

4.1.1 Starting the Installation using Recovery DVDs

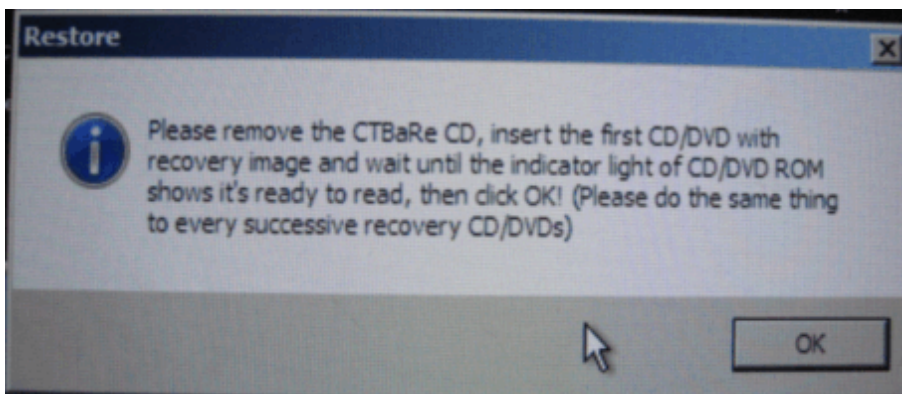
- Along with the IES Tower model, select the correct CD: ***CD_SOM/5_ICSI/IES_VB42B_32bit_BARE*** or ***CD_SOM/5_IES_VB42B_64bit_BARE***, and insert it into the CD-drive.
- Select **System> End> Restart System** and boot from the CD.
- Disconnect the Ethernet cable of the IES.
- Wait until the **Backup/Restore** window appears, select **Restore System** and press **OK**.
 - » The following message appears, press **Yes**.

Fig. 22:



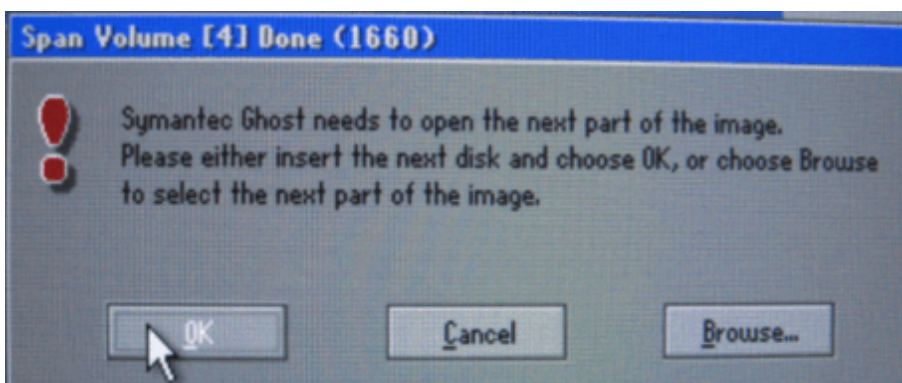
- The following message appears: Please remove the CTBaRe CD, insert the first DVD with the recovery image and ... CD/DVDs).

Fig. 23:



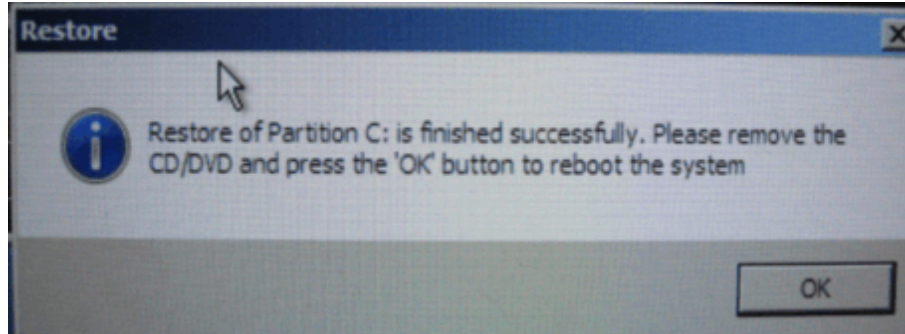
- Insert the corresponding recovery DVD. When the DVD is ready, press **OK**.
- The system drive is being restored. This will take about 10 mins.
- Insert the 2nd DVD when you are asked. Click **OK** when the DVD is ready.

Fig. 24:



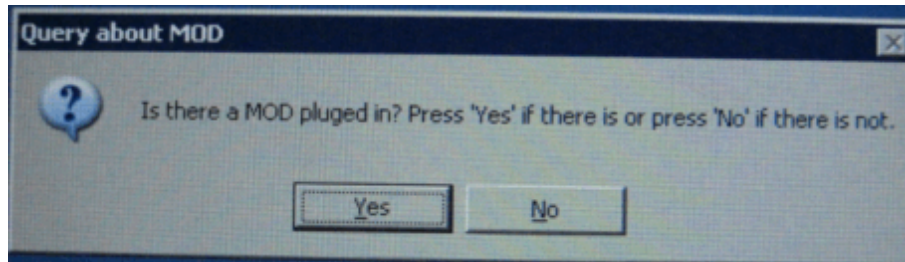
- When the restore process is finished, the message appears: *Restore of Partition C: is finished successfully...* Remove the DVD and click **OK** to reboot the system.

Fig. 25:



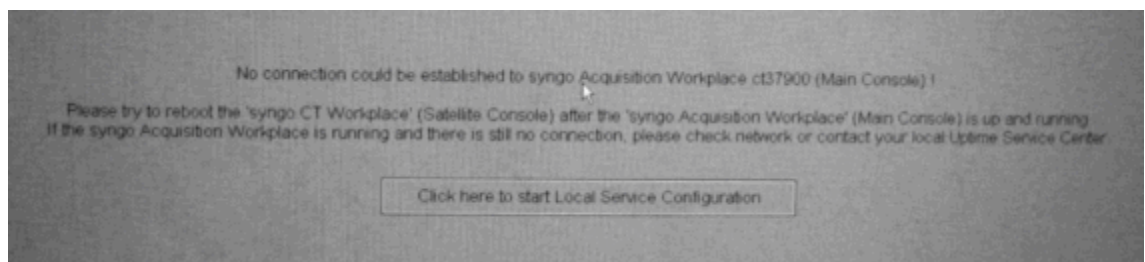
- Wait until the message: *Is there MOD plugged in?...* appears, click **Yes** or **No** according to whether there is a MOD connected with the IES.

Fig. 26:



- » The system will check the partitions.
- Wait for about 5 min., the system will reboot automatically.
- Enter the service software by clicking **Click here to start Local Service Configuration** and go to the Restoring Steps(→ Restoring Steps / Page 49).

Fig. 27:



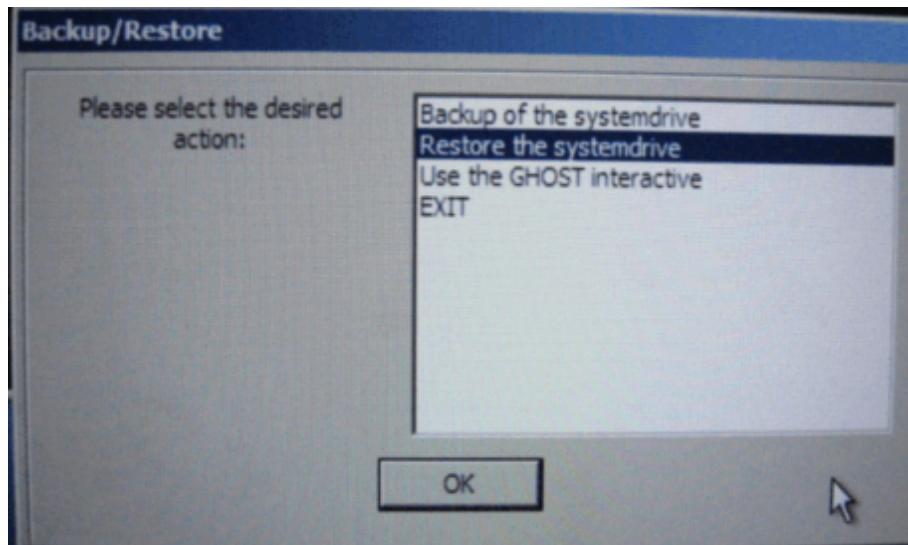
4.1.2 Starting the Installation using the delivered VB42B DVDs



If the IES has a MOD connected, you need to remove the MOD from the system (disconnect the MOD drive) and then begin the SW installation. After the SW installation, the MOD drive has to be reconnected.

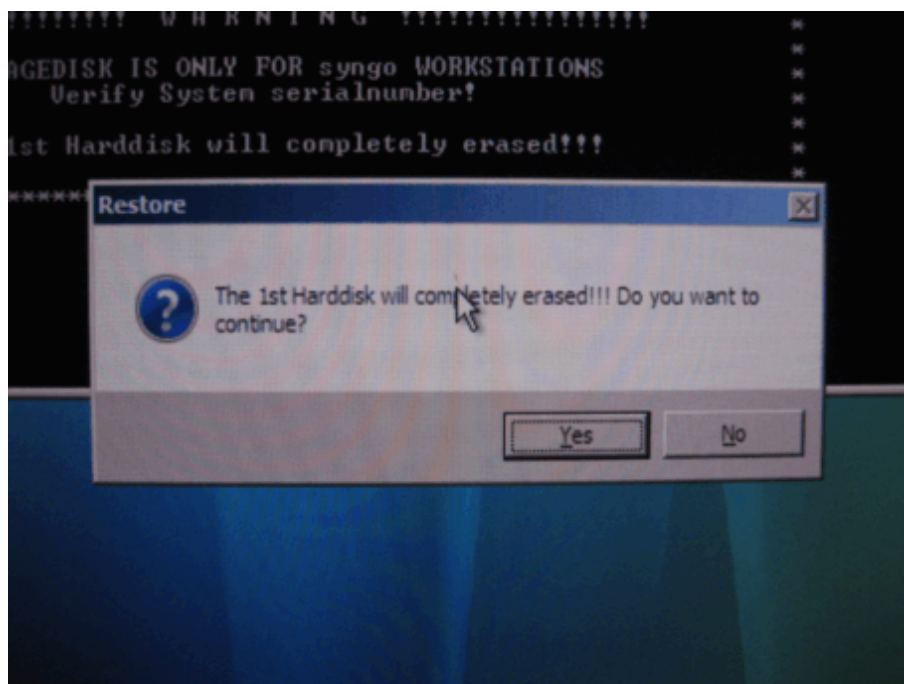
- Insert the DVD_Som/5_ICS/IES_VB42B_1_of_2_32bit or DVD_SOM/5_IES_VB42B_1_of_2_64bit into the IES.
- At the IES, select **System > End** and reboot the IES.
- Disconnect the Ethernet cable of the IES.
- The computer boots from CD-ROM. After a short while, the message "Windows is loading files" appears. Wait until a Backup/Restore window shows up.
 - » Select **Restore the system drive** using the mouse or Tab key and arrow keys on the keyboard, and click **OK**.

Fig. 28:



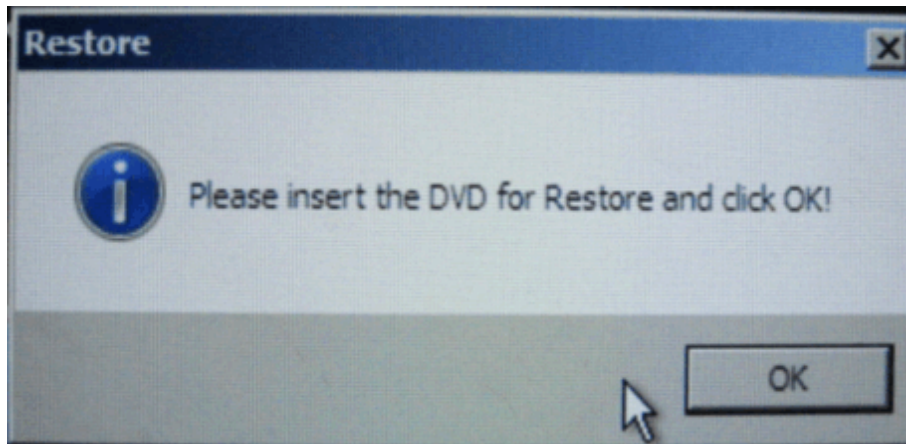
- A message 'The 1st Harddisk will be completely erased!! Do you want to continue?' pops up. Click **Yes**.

Fig. 29:



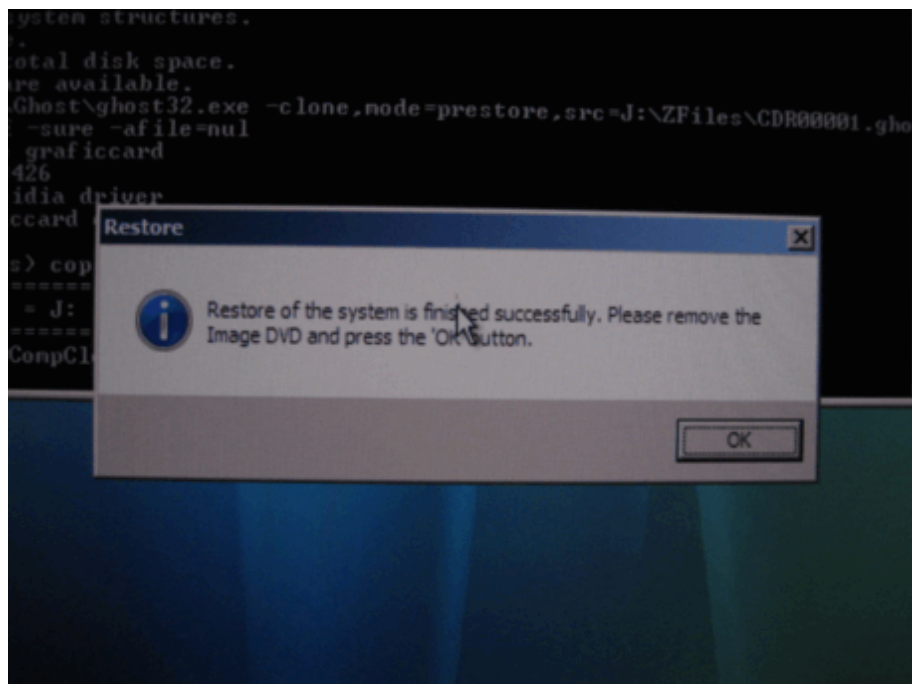
- A message Please insert the DVD for Restore and click OK. pops up. Click **OK**.

Fig. 30:



- » After the disk drives check, the GHOST is running. The system is copying files.
- Insert **DVD_SOM/5_ICS/IES_VB42B_2_of_2_32bit** or **DVD_SOM/5_IES_VB42B_2_of_2_64bit** when you are asked , click **OK** when the DVD is inserted and ready.
- When the Restore process is finished, a message Restore of the system is successfully... appears. Remove the DVD and click **OK**.

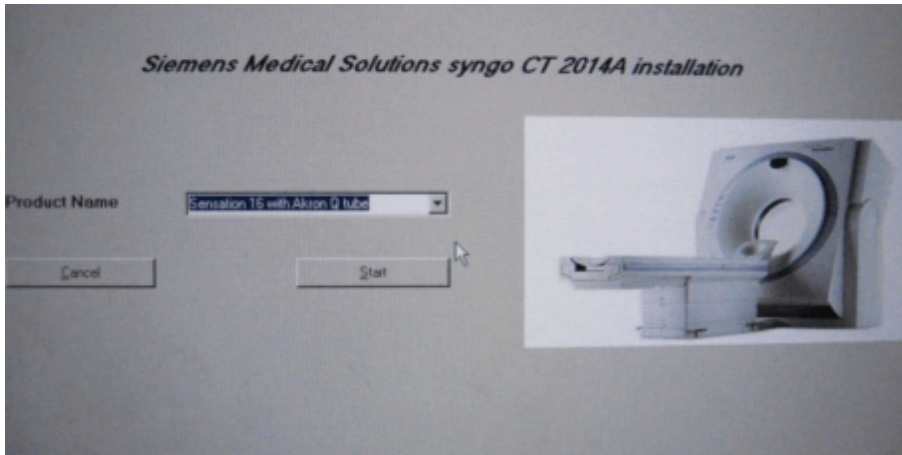
Fig. 31:



- The system restarts automatically and continues with the installation.
- The system will reboot 2 times.

- Wait until the following screen appears (the picture is only an example).

Fig. 32:



- Select **syngo CT Workplace** in the **Product Name** field.

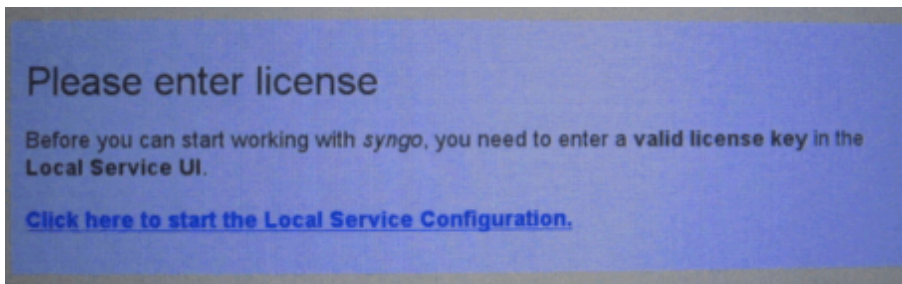
Fig. 33: (This is only an example)



- Click **Start**.

Completion of IES installation (approx. 20 minutes) is indicated by a pop-up window with the message **Please enter license**.

Fig. 34:



- Click **Click here to start the Local Service Configuration** to start the service software (complete service key and system serial number are required).

4.2 Restoring Steps



Restoring steps have to be implemented after the software installation, using the Recovery DVDs or original software DVDs.

4.2.1 Restore

- In the main menu of the service software, select the **Backup & Restore** function.
- Insert the **IES** backup CD.
- Under **Command**, select **Restore**.
- Under **Drives**, select the corresponding drive.
- Under **Archive**, select **SW-Settings02-XXX**.
- Under Groups, select all and click **Go**.
 - » `BackupRestore Running . . .`; wait until `Ready` is displayed.



Ignore the error ".....C:\Somaris\log\FilmRestore.log for more detailsUpdate.bat finished-File not found - Layout".



Ignore the error ".....The hostname of current machine is different from the hostname in the security settings backup.

- Under **Archive**, select **CT-Settings-XXX**.
- Under **Groups**, select all items and click **Go**.
 - » `BackupRestore Running . . .`; wait until `Ready` is displayed.



Ignore the error "c:\somalislutils\ImageTxtAdaption.pl": No such file or directory.

- Under **Archive**, select **SecuritySettings-XXX**
- Under **Groups**, select **all** and click **Go**.
 - » `BackupRestore Running . . .`; wait until `Ready` is displayed.



Ignore the error ".....The host name of the current machine is different from the host name in the security settings backup."

- Under **Archive**, select **InSpace-xxx** (if available) and click **Go**.
 - » `Backup Restore Running . . .`, wait until `Ready` is displayed.
- Take out the backup CD.
- Reconnect the Ethernet cable.



Make sure the ICS is ready and running.

- Click **Home**, and confirm the system reboot.



If a problem exists in connecting with the ICS, restart the ICS with the IES.

4.2.2 Configuration Walk-Through

- In the main menu of the service software, select **Configuration**.
If you see the Initial setup page, perform the following steps.
- Click **Next** on the **Initial setup** page
- Set the Administrator password to the default string.



This password should be set to the default string, otherwise, the “SolidCore startup failed” window will show up when the system starts.

- Click **Next** or **>** for all pages in the later pages, (for SRS, set according to the previous record and save), until the **Finish** button appears at the bottom of the page.
- Click **Save** (if applicable) and then **Finish** to save the settings.
- Click **Home** and if you have changed something, you may be required to reboot the system, confirm the rebooting. If there is no rebooting demand, close the service window.

4.3 Completion of Configuration

4.3.1 Tuning CD Burning

- In the service software menu, select **Utilities**.
- Under **Source**, select **Tune CD Burning**.
- Activate the **Tune CD Burning for the current hardware** checkbox.
- Activate with **Return** or **Go**.
- Wait until the message `Drives have been successfully installed` appears, and click **OK**.
- Click **Home**.
- Close the service window.

4.3.2 Remote Service Access

- In the syngo user menu, select **Options > Service > Remote Service**.
- Check the option according to the previous record.
- Click **OK** to close the window.

4.3.3 Copying back user-specific Data

Open Windows Explorer and copy back the user specific data from the storage medium to the **H:\Sitedata** directory. When asked for "confirm folder replace", click the **Yes to All** button.

4.3.4 Importing Reference Images (optional)

- From the Patient Browser, select **Transfer > Import from Offline**.
- From the drop-down list, select the **H:\SiteData\ReferenceImages** path.
- Select all images and click **OK**.
 - » After the import, the images are available in the Patient Browser.

4.3.5 Configuring the Keyboard Language

Perform the following steps to configure keyboard languages other than English (**except for Chinese and Japanese**):

- From the syngo user interface, select **Options > Configuration**.
- In the configuration panel, double-click the **Regional and Language Options** icon.
- Set the proper language and location.
- Click **Customize...** and set the time and date format according to the previous record and click **OK**.

- Select the **Languages** tab card, set the language to be used in the menu and dialogue windows and then click **Details**.
- Click **Add**, select the correct input language and confirm with **OK**.
- Select the English keyboard language, click **Remove**, and confirm with **OK**.
- Click **OK** to close the window.
- Click **OK** to confirm the change.
- Confirm restart of the system.

Attention: For **Chinese/Japanese**: Open the **DICOM/Character Set** dialog in Local Service. Select the corresponding character set, for Japanese "**Japanese, Katakana**" and check the option **Allow character set extension**; for Chinese **Unicode** or **Chinese**. Click **Save** and click **OK** to confirm the saving. Click **Home** and confirm the application restart.



If a blank web page appears for some time, the computer has to be restarted manually.

4.4 Known Bugs and Errors

The following points are known issues and need manual intervention.

1. For dual monitors on the ICS or IES, the syngo task bar will be displayed at a higher position than normally. As a result, the relevant parts of the syngo User Interface are not visible. This can be corrected as follows (under the administrator's account):
 - Open **Start->Settings->Control Panel->nVIDIA nView Desktop Manager**.
 - Deactivate the nView Desktop Manager.
 - Restart the system.
 - After restarting the system, the failed task bar should be fixed.
2. The MIP rendering algorithm under **Local Service > Configuration > Application** has to be set manually, since it depends on the system's hardware.

5.1 Software Self-Test (ICS and IES)

- Open the **Local Service Home** menu and go to **Test Tools**.
- Click the **Software SelfTest** and execute all subtests.
 - » In case of errors, refer to the hints for troubleshooting.

5.2 SRS Configuration (ICS and IES)



Beginning with Somaris/5 VB40, the System Management Software CA Unicenter was replaced with HP OpenView. HP OpenView uses different IP addresses and 2 new ports that have to be enabled in the customer IT environment. If the System Management configuration fails, please refer to document CT00-000.843.08.xx.02 (SRS configuration) .

- Open the local service home menu and go to **Configuration**
- Click **System Options**
- Check **System Management** under the List of System options, if not already activated.
- To complete the initial setup, click **Next** or **>**.
- At **Local Host > Site info**
 - Check / select your country
 - Check / select the Remote Service Center used
 - Check / select the correct System - Serial no.(Serial no."0" must not be entered)



If the serial number has been changed, System Management has to be uninstalled first and reactivated afterward.

- Click **Finish** or **Save** if you have made any changes
- Go to **Service >SRS**, check/enter the SRS settings and perform the SRS configuration



Outside NAT: If outside NAT was configured before, it will not be restored correctly. The new HP OpenView software uses only two servers which have to be configured now.

- HP software distribution server: Enter the IP address of the CS Asset Server
- HP Event Management: A new IP is required. Do not use any of the old CA IPs!



Static Route to SRS: If a static route was configured before, it will be restored automatically. No action by the CSE is necessary.



SRS function: SRS Base, HTTPS proxy (NetViewer, CSKB), and System Management will always be configured.

Virus Protection : Enable the checkbox if the customer used this service before.

Utilization Management: Enable the checkbox if it was configured before.

- Click **Save**. All SRS functions will be configured automatically. The message "Site data successfully saved" is output. Click **OK** to confirm.
- Click **>** to make the SRS function check.
 - » The SRS function check will be initiated.

- After checking, click **Finish** and **Home** to return to the service menu.

5.3 Set up of Auto Reports (ICS and IES)



Neglect the contents not available in IES.

5.3.1 General Information

The SRS connectivity is a cornerstone in maximizing the uptime of Healthcare systems and thus increasing customer satisfaction and serviceability.

Therefore, every system is typically connected to SRS, using this connection for RSC/HSC support, for the distribution of updates, and for application support (where available).

To improve the quality and the reliability of the products with SRS connection, the auto report setup is becoming mandatory.

In this section, the (predefined) **Mon_HQ** auto report is activated and a tube data report is included in the auto reporting setup.

Following these instructions is mandatory. Only in exceptional cases where the contract does not allow sending this data is it permissible to skip this section.

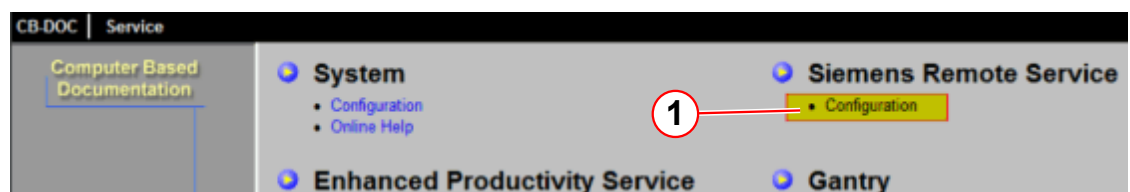
If no auto reports are received after this update has been performed, Siemens reserves the right to follow up. If no credible reason is given for not enabling the auto report, the update status may be reset to "open".

5.3.2 Setting up Auto Reports

1. Check the general configurations of the SRS connection.

For detailed instructions on the setup and configuration of the connectivity, consult the CB Doc.

Fig. 35: For details on the SRS setup, refer to the CB DOC



(1) Section SRS configuration in the CB DOC

(the setup of the auto reports will be adapted in future revisions of the SRS configuration document to correspond to this description)

2. Go to **Configuration > Service > Auto Transfers** and activate the checkbox **activate automatic Transfers**.

Fig. 36: Check "activate automatic Transfers"



(1) Select the menu item "Auto transfers"

(2) Check "activate auto Transfers"

3. Select the configuration item **Escalation** and enable the checkbox(es) under **Additional Data Configuration**

The layout differs marginally in SOMARIS 5 and SOMARIS 7, see the screenshots.

Fig. 37: Check the box "Generate tube history (HUT) files"

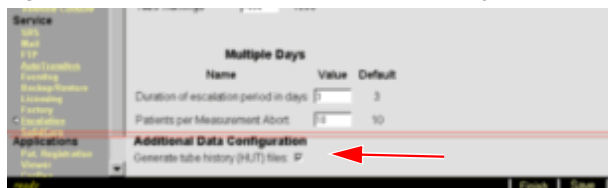
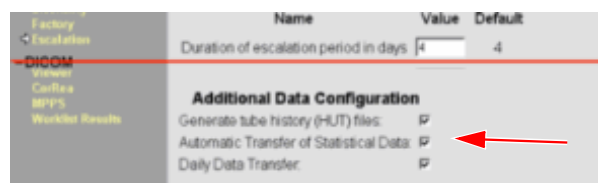
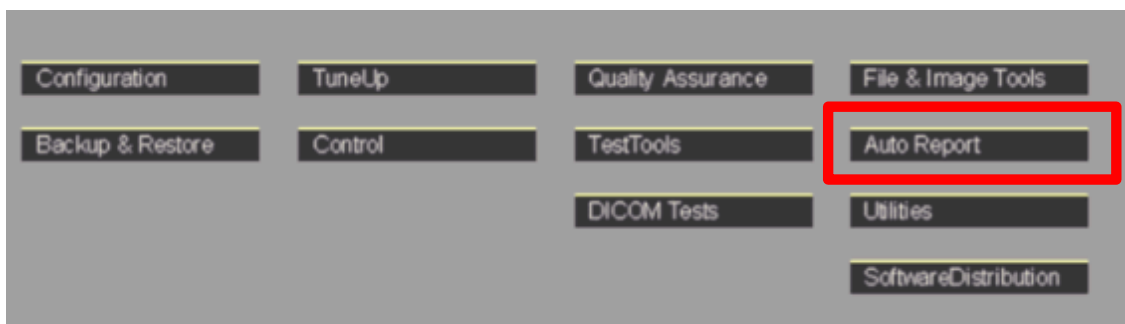


Fig. 38: Activate all checkboxes under "Additional Data Configuration"



4. Go to the Service menu, section **Auto Report**

Fig. 39: Go to the service menu "Auto Report"



(1) AutoReport

5. To create the new auto report select **TEMPLATE** (only in ICS)

Fig. 40: Empty auto report template

- (1) Select TEMPLATE
(2) Create the name "hut_transfer"

Set up the report using the following parameters:

Fig. 41: Create the auto report "hut_transfer"

- (1) Active: checked, name: hut_transfer, target: srs_mail, start date: today, end date: 2030
(2) Sample source: File
(3) Sample what? see list below
(4) Sample when? see list below
(5) Send what? see list below
(6) Send when? at once

Tab. 1 Parameters for auto report "hut_transfer"

Number	Parameter	Configuration
1	Active Name Target Begin, End	checked hut_transfer srs_mail today, 2030
2	Source	File
3	Sample what?	File name: C:\Somaris\service\extract\performance*.hut Att.: type exactly as shown: no spaces Mode: time triggered Search pattern: empty, lines:1
4	Sample when?	continuously Sample times: 02:00;07:00;08:30;10:00 (you may copy this setting from the predefined "Mon_HQ" report)
8	Send what?	Mail: as separate files FTP target: default target as set up in the SRS configuration ("ftp2lux09505" was used for the screenshot of a German system)
9	Send when?	at once

6. Click **Save** and confirm the prompted window to save the change.
7. Activate the pre-defined auto report **Mon_HQ**.

Fig. 42: Auto Report "Mon_HQ"

The screenshot shows the 'Auto Report' configuration window for the 'Mon_HQ' report. The window has a title bar with 'Service Key Expiry Date: 17-Sep-2014' and navigation buttons 'Home', 'Event Log', and 'Help'. The main area is divided into several sections:

- REPORT:** A dropdown menu showing 'Mon_HQ'.
- Active:** A checkbox that is checked.
- Name:** A text field containing 'Mon_HQ'.
- Target:** A dropdown menu showing 'srs_mail'.
- Begin:** A date/time picker set to '31' of 'October' '2001'.
- End:** A date/time picker set to '31' of 'October' '2030'.
- Enabling Status:** A list of reports with checkboxes: 'Mon_HQ' (checked), 'Tube_extract' (unchecked), and 'Tube_history' (unchecked).
- Sample source?:** A dropdown menu showing 'File'.
- Sample what?:**
 - Filename:** A text field containing 'extract\Evt_20.txt'.
 - Search Pattern:** An empty text field.
 - Mode:** A dropdown menu showing 'Time Triggered'.
 - Lines:** A dropdown menu showing '1'.
- Sample when?:**
 - Frequency:** Radio buttons for 'Continuously' (selected), 'Daily', 'Weekly', and 'Monthly'.
 - Sample times:** A text field containing '02:00;07:00;08:30;10:00'.
- Send what?:**
 - Mail:** A dropdown menu showing 'as separate file(s) without mails'.
 - FTP target:** A dropdown menu showing 'ftp2lux09505'.
 - Enable file compression:** An unchecked checkbox.
 - Command:** An empty text field.
- Send when?:** Radio buttons for 'At once' (selected), 'Daily', 'Weekly', and 'Monthly'.

At the bottom right, there are buttons for 'Save' and 'Delete'.

8. Click **Save** and confirm the prompted window to save the change.

5.4 Final Check (ICS)

- Open the Local Service Home menu and go to **Configuration**.
- Click **Technical** and >.
- Go to **Local Host > Technical**.
- Select **Country for Quality**.
- Under **Constancy**, select the value along the record before.



Under the Constancy item, there is a new **"Included"** list. Please activate the related checkbox along local regulations.

Fig. 43: It's only an example

Configuration

System Options

- Local Host
 - Site Info
 - TCP/IP LAN
 - Application
 - ➔ Technical
 - Serial
 - Version
 - Users
 - MonitorType
- Security
 - Settings
- Service
 - SRS
 - Mail
 - FTP
 - AutoTransfers
 - Eventlog
 - Backup/Restore
 - Licensing
 - Factory
 - Escalation
- DICOM
 - General
 - Character Set
 - Structured Reports
- System Mgmt
 - Master
 - Agent Controls
- Applications
 - Pat. Registration
 - ...

Quality Configuration

Language for Quality PDF Protocols

Country for Quality

Constancy

	Included	Expired	Out of tolerance
Calibration	☒	☐	☐
Phantom Check	☒	☐	☐
Lightmarker	☒	☒	☒
Sag/Cor. Lightm.	☒	☒	☒
Prev. Image Pos.	☒	☒	☒
Slice	☒	☒	☒
Homogeneity/Water	☒	☒	☒
Noise	☒	☒	☒
MTF	☒	☒	☒
Contrast	☒	☐	☐
Low Contrast	☒	☐	☐
Table Position	☒	☒	☒
CTDI Air	☒	☐	☐
CTDIw Head	☐	☐	☐
CTDIw Body	☐	☐	☐

- Click **Save** and **OK** in the window that is displayed.

- Click **Home**.
- In the Local Service Home menu, go to **Quality Assurance**.
- Click **Constancy**.
- Select **Reference** (click the icon next to **Constancy**).
- Click **Go** and proceed according to the program instructions.
- Open the Local Service Home menu and go to **TestTools**.
- Select **Completion Test**.
- Click **Go** and proceed according to the program instructions.
- Perform scans in the customer mode and expose a film to ensure the camera is functioning properly.

5.5 E-Manual CD Installation and Checking



The E-manual is a collection of books provided directly on the ICS/IES. The books provide detailed information about the CT system.

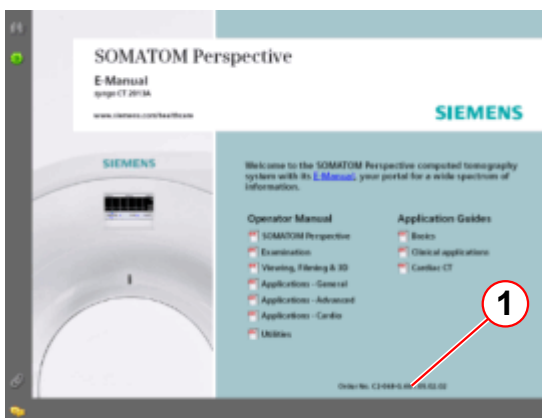
5.5.1 E-Manual Installation

- Insert the E-manual CD
- From the syngo user interface, select **Options > Service > Local Service**.
- Select **Utilities**.
- Under **Source**, select **Escape to OS** and under **Command**, select **NT Command (Secure)**.
- Under **Parameters**, type in **R:\install.bat** ↵
- Activate with **Enter** or **Go**.
- Confirm the prompted window.
- Wait until the message `Installation is finished successfully` appears.
- Close the service window.
- Take out the E-manual CD and restart the system

5.5.2 E-Manual Checking

- From the syngo user interface, select **Help > E-Manual**
- The following index page appears.

Fig. 44: E-Manual Cover Page (example only)



(1) E-Manual print no.

- Check if the print no. is same as that on the E-manual CD.

5.6 Backup at the ICS and IES

5.6.1 CD Write Check

During previous installations, we've experienced problems writing the backup files to CD. As a result, this check has to be executed before creating the backup CD.

- Open the local service home menu and go to **Configuration > Offline Devices**.
- Check the offline device.
 - » Depending on the offline device installed, in the drop-down menu select either **CD-R** or **CD/DVD Combo** as an offline device.
 - » Check the item: **Default device**. Make sure the recording speed is set to **Half Speed**, or a recording error may occur.

5.6.2 Writing the Backup CD

- Start the service software and select the **Backup & Restore** function in the main menu.
- Insert the backup CD-R for the **ICS or IES**.
- Under **Command**, select **Backup**.
- Under **Drives**, select the CD drive, e.g., [-R-] CD-R.
- Under **Packages**, select **CT-Settings**, **SW-Settings02**, **Security-Settings**, **InSpace** and **SRView** (if applicable), respectively and click **Go**.
 - » BackupRestore Running . . . ; wait until Ready is displayed.
 - » Different backup files will be created.
- Remove the backup medium and mark the CD with proper information, such as workstation type, serial number, and date etc.
- Click **Home** and terminate the service software by closing the Browser window.

5.7 Creating a Recovery DVD for ICS and IES



In the following context, the installation CD/DVD of VB42B is mentioned. When VB42B_SP1 is installed, the corresponding VB42B_SP1 CD/DVD should be used instead.



A Recovery DVD has to be created only if the delivered VB42B software DVDs were used for the installation.

After the recovery DVD has been created, it has to be checked.



Pay attention to the procedures below, they are different from the procedures used previously. Otherwise, the newly generated DVDs cannot recover the system.

Create a recovery DVD

- Insert the correct CD:
 - For IES Tower 8 or lower, e.g. Tower 7 or 7*, and all ICS Towers, select **CD_SOM/5_ICS/IES_VB42B_32bit_BARE**.
 - For IES Tower 8D/E and higher, such as Tower 10D/E, Tower 11D/E, select **CD_SOM/5_IES_VB42A_64bit_BARE**.
- In the service software menu, select **Utilities**.
- Under **Source**, select **Escape to OS**.
- Under **Command**, select **NT Command Interpreter**.
- Under **Parameters**, type in **R:\PRE4BARE.bat** ↵ and activate with **GO** or **Enter**.
- The system will make some preparations in SolidCore and then reboot from the CD.
- After the system rebooted successfully, select **Backup system**, and press **OK**.
- When asked for a blank DVD, remove the CD and insert an empty DVD. When the DVD is ready, press **OK**.
 - » An image of the C: partition is created.
- Insert the 2nd DVD when you are asked. When the DVD is ready, click **OK**.
- When the recovery process is finished, remove the DVD and press **OK** to reboot the system.
- Label the recovery DVDs with the following information: Tower part no., revision no., software version, system serial no. and so on.

5.8 Checking the Recovery DVD for ICS and IES



In the following context, the installation CD/DVD of VB42B is mentioned. When VB42B_SP1 is installed, the corresponding VB42B_SP1 CD/DVD should be used instead.

- Insert the correct CD:
 - For IES Tower 8 or lower, e.g. Tower7 or 7*, and all ICS Towers, select **CD_SOM/5_ICS/IES_VB42B_32bit_BARE**.
 - For IES Tower 8D/E or higher, such as Tower 10D/E, Tower 11D/E, select **CD_SOM/5_IES_VB42B_64bit_BARE**.
- Select **System> End> Restart System** and boot from the CD.
- After the system rebooted successfully, select **Check recovery DVD**, and press **OK**.
- Insert the first recovery DVD. When the DVD is ready, click **OK**.
- The Ghost window appears. The system is checking the first recovery DVD. Insert the 2nd recovery DVD when you are asked.
 - If the message **The recovery DVD is OK! Please ...** appears, it means that the recovery DVDs are OK.
 - If the message **Cannot read from file & The recovery DVD has some errors!...** appears, you need to create new recovery DVDs.
- Take out the recovery DVD and click **OK**. The system will reboot automatically.
- Leave the Recovery DVDs at the customer site.

5.9 Customer Information regarding Expert-i



Expert-i is a software that allows the customer to access the ICS/IES from a remote client. The required software is already installed on the ICS/IES. Viewer installation from the Expert-i CD at the remote client is performed by the customer. The installation for the client PC is described in the syngo Expert-i Operator Manual.



There are currently 2 versions of the Expert-I client, VA10A and VA20A. VA10A is for Somaris VB30B, VB38B and VB39B, while VA20A for VB40B, VB41B and VB42B.

- Hand over the Expert-i CD (VA20A) and the syngo Expert-i Operator Manual to the customer.

Provide the following to the customer:

- IP address of the ICS/IES.
- Host name of the ICS/IES.
- CD with client software.

6.1 SRS Troubleshooting and Utilization Management

6.1.1 SRS Troubleshooting

In most cases the SRS configuration fails with the SRS function: **System Management**. This section provides information only for System Management.

Additional SRS troubleshooting information is available under "SRS Configuration", print no. CT00-000.843.08.04.02. The document is part of the CB-DOC for the Somatom Definition systems.

6.1.2 Troubleshooting System Management

Check the result of the configuration log file

Each System Management configuration attempt writes a log file stored under C: \sysmgmt\log\g.**sysmgmt_install.log**.



Check in the log file for the first E = Error in the far left column; it describes the reason for the failed installation. Perform the corrective action for the error type.

Tab. 2

Error type	Corrective action
E 08/05/2008 15:45:14 Failed reaching Management Server e.g. 194.138.39.19 Please check Network settings and try again	Refer to the section "Known errors in Somaris/5 VB40 and up"
E 08/05/2008 15:45:14 Failed to get Backend Data	The registration data could not be retrieved from the SRS server. Contact the SRS Help Desk for support.
E 08/05/2008 15:45:14 Failed to Install Certificate of Event Agent	Repeat the System Management configuration once. If the error persists, contact the SRS Help Desk for support.
E 08/05/2008 15:45:14 Failed Installation of Template	Repeat the System Management configuration once. If the error persists, contact the SRS Help Desk for support.

6.1.3 Known Errors in Somaris/5 VB40 and up

Two possible problems have been noticed in Somaris/5 VB40 and up

- The system cannot reach the Management Server because the HP OpenView proprietary ports 12061 and/or 13001 are not enabled in the customer firewall, or the corresponding HP Event Management IP address is not routed correctly. The SRS server-specific Event Management IP address is
 - Europe/Furth:194.138.39.19

- Asia/Singapore:194.138.243.179
- America/Newark: 129.73.116.91

» Contact the customer network administrator or the SRS Help Desk for support.

- The system is using Outside NAT. The restored NAT addresses from VB20/VB30 are incorrect in Somaris/5 VB40. As a result, the Management Server cannot be reached. Instead of connecting to one of the above IP addresses, the system tries to reach a NAT address.

Example: E 08/05/2008 15:45:14 Failed reaching Management Server **e.g. 213.189.103.5!** Please check Network settings and try again.

- » On the SRS page, enable the **Outside NAT** checkbox and enter the correct addresses. The 3 NAT addresses are provided by the network administrator or the person responsible at Siemens.

Tab. 3

SRS servers	Real IP addresses	NAT addresses to be entered under Outside NAT
SRS Access Server IP	Eu- rope=194.138.38.18Asia=1 94.138.243.178Ameri- ca=129.73.116.92	This NAT address is restored correctly by Backup/Restore
HP OpenViewEvent Server IP(*)	Eu- rope=194.138.38.19Asia=1 94.138.243.179Ameri- ca=129.73.116.91	IT Admin or the person responsible at Siemens provides a new NAT address for the HP Event Management server
HP OpenView Distribution Server IP(**)	Eu- rope=194.138.38.21Asia=1 94.138.243.181Ameri- ca=129.73.116.93	Enter the former CA Asset Server NAT address here.

(*).... The HP Event Management server is still the same as that known from the former Magic-Watch function used in Somaris/5 VA40-VA70.

(**). The CA Unicenter Asset server was integrated into the CA Software Distribution server. The former IP address of the CA Asset server is now used by the HP Software Distribution server.

7.1 Changes to previous version

Chapter	Change	Reason
Chapters 1 and 2	Incorrect directory path in the notes corrected.	ECR 623783
Chapters 2 and 3:	Note modified, "Tower selection table" added.	ECR 627350

There are no Hazard IDs in this document.

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